



EAST WEST UNIVERSITY

Department of Computer Science and Engineering

Assignment

Course Code: CSE495

**SkillMatch: A Platform Connecting University
Students with Freelance Projects**

Submitted To

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Submitted By

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Introduction

SkillMatch is a freelancing platform designed to bridge the gap between university students and startups. It empowers students by providing real-world project experience while enabling businesses to access affordable, skilled talent for short-term tasks.

Problem, Market Need, and Solution

University students often struggle to gain real-world experience, while startups and small businesses face a shortage of affordable, skilled help for short-term projects. Currently, no platform effectively connects students with micro-jobs that match their skills. **SkillMatch** addresses this gap by linking students to freelance projects, enabling them to build experience and earn income, while helping businesses complete tasks efficiently and cost-effectively.

Project Objectives from a SMART Perspective

From a SMART perspective, the SkillMatch project aims to connect students with projects (Specific), measure success by achieving 50 student profiles and 20 active jobs (Measurable), implement a pilot using university resources (Achievable), ensure relevance by supporting both students and startups (Relevant), and deliver a minimum viable product within five months (Time-Bound). This structured approach ensures clear, trackable, and achievable project goals.

Key Project Management Components

Effective project management ensures SkillMatch is completed on time, within budget, and meets its objectives. Key components like the project charter, WBS, Gantt chart, network diagram, and costing provide structure, clarity, and control throughout the project.

1. Project Charter

- a. Defines project purpose clearly.
- b. Authorizes projects and resources.

2. Work Breakdown Structure (WBS)

- a. Breaks the project into smaller tasks.
- b. Provides clear task hierarchy.

3. Gantt Chart

- a. Visualizes project timeline easily.
- b. Shows task dependencies and duration.

4. Network Diagram

- a. Illustrates task sequence logically.
- b. Highlights project critical path.

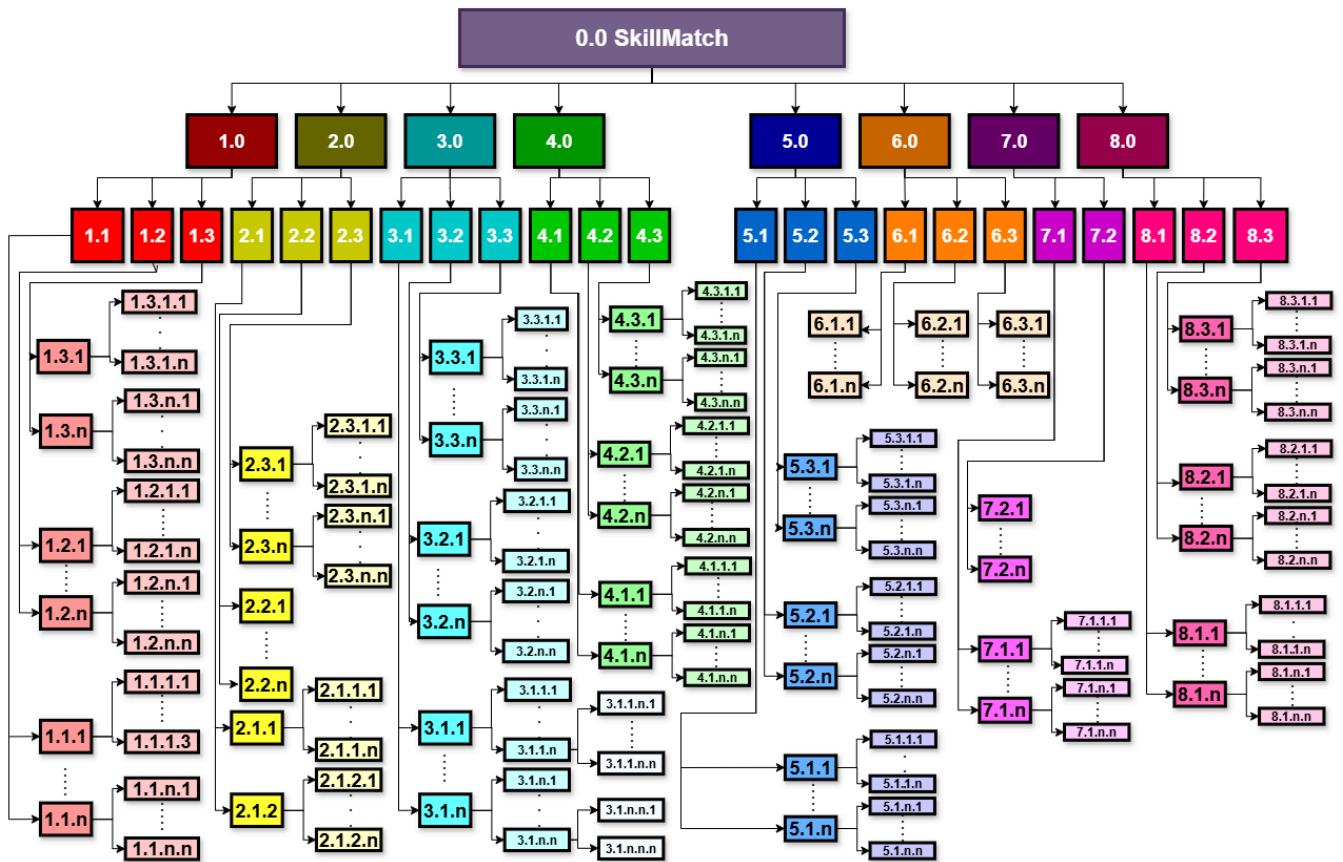
5. Costing

- a. Estimates project financial requirements.
- b. Tracks budget allocation and usage.

1. Project Charter

Project Charter								
Project Title		SkillMatch: A Platform Connecting University Students with Freelance Opportunities						
Project Purpose		SkillMatch links students with freelance micro-projects to gain experience and earn, while startups access affordable skilled talent for short-term tasks.						
Project Approval		Prepared By :	1. Maria Hasan	2. Shamima Sharmin Ananna	Approved By :	Deepak Chandra Ray (Course Instructor)	Approved Date	20/7/25
			3. Jerin Anan Proma	4. Fatema Tuz Zannat				
Project Objectives			Milestones		Time	Deliverables		
1. Develop MVP freelance web platform			1. Project Planning & Charter Approval		21days(9/1/25)	Requirements, WBS, Timeline		
2. Enable student registration and profiling			2. Platform Design Completion		20 days	UI/UX, Branding , Architecture		
3. Implement job browsing and filtering			3. Core Development Completion		40 days	Functional modules, Admin Panel		
4. Integrate communication and admin tools			4. Testing & Debugging		41 days	Unit, Integration user acceptance testing		
5. Achieve pilot adoption and satisfaction			5. Pilot Launch & Evaluation		28 days(4/1/26)	Hosting, Feedback, Data analysis		
Project Teams			Sponsors		Budget Estimate			
1. Project Manager			1. East West University Career Center		Category		Estimated Cost (BDT)	
2. System Analyst			2. Bangladesh Youth Foundation		1. Development		~74.4% ≈ 21,083 USD	
3. Front-End Developers			3.Grameenphone Innovation Fund		2. Management		~7.1% ≈ 2,000 USD	
4. Back-End Developers			4. Startup Bangladesh Limited		3. General Labor		~18.5% ≈ 5,250 USD	
5. QA Tester			4. Orange Corners Bangladesh		Total		28,333 USD	
6. UI/UX Designer			Constraints		Limited technical resources or skills may delay development, while low student or employer adoption and technical issues during the pilot phase could affect usability.			
7. Marketing/Outreach								

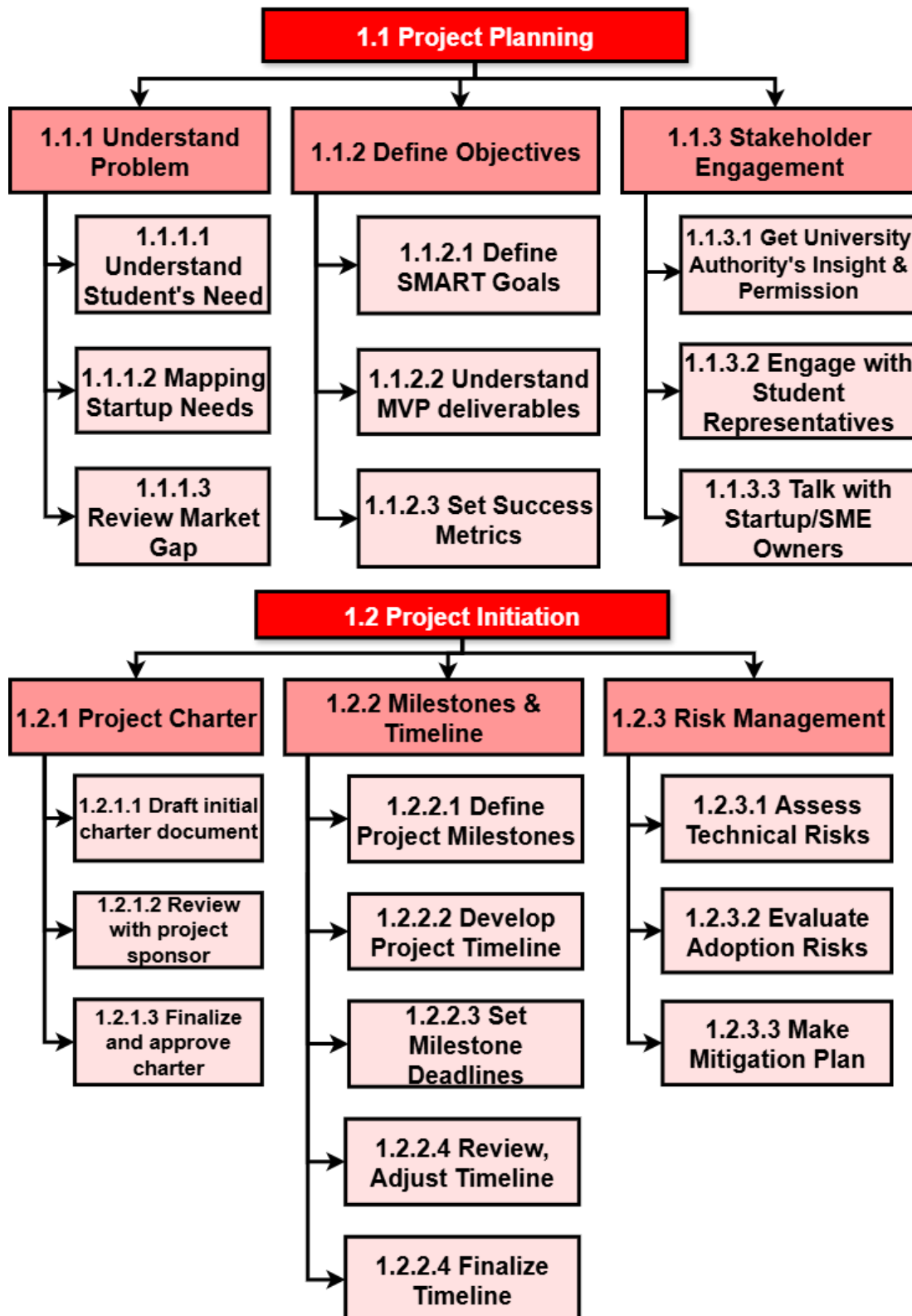
2. Work Breakdown Structure

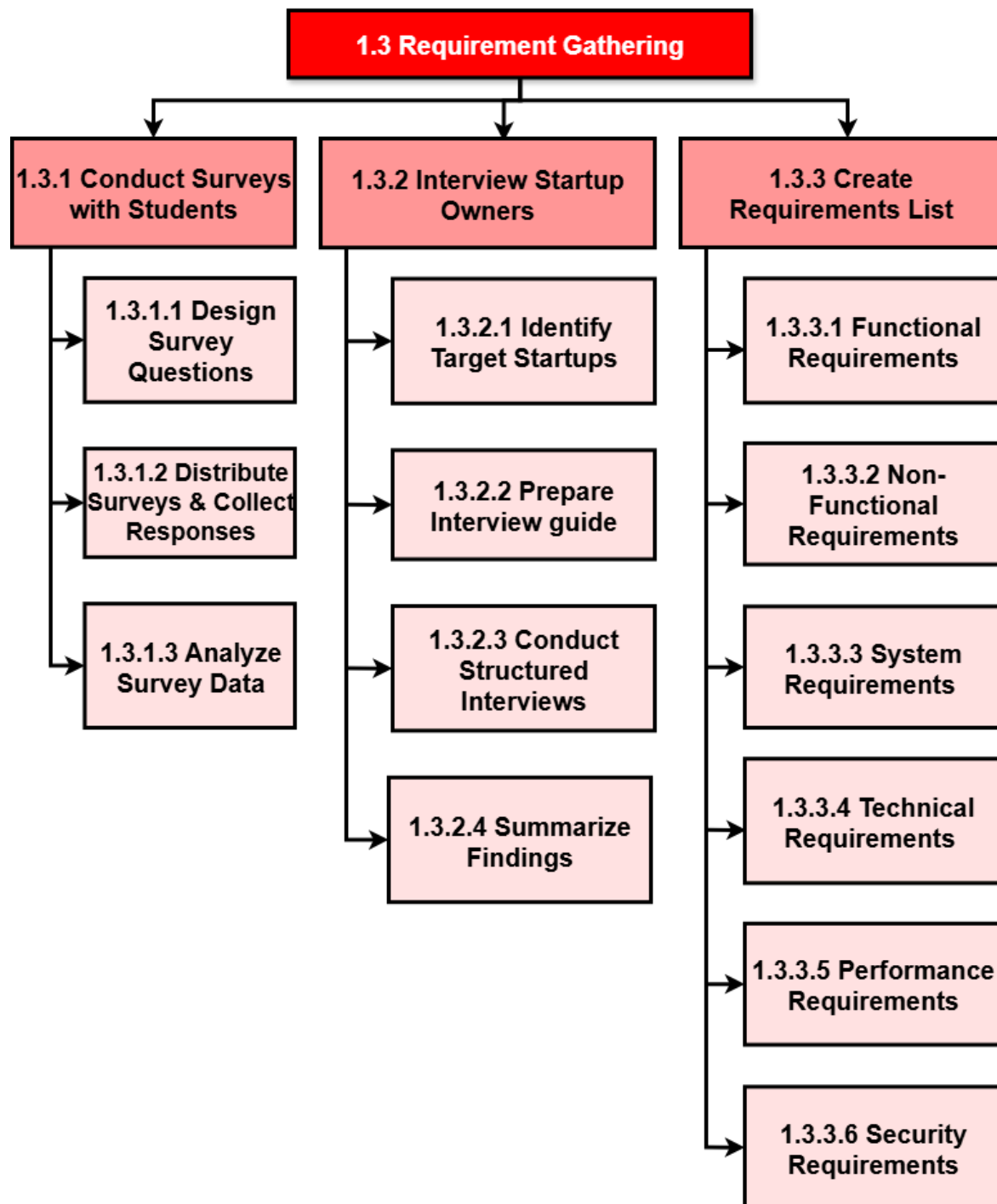


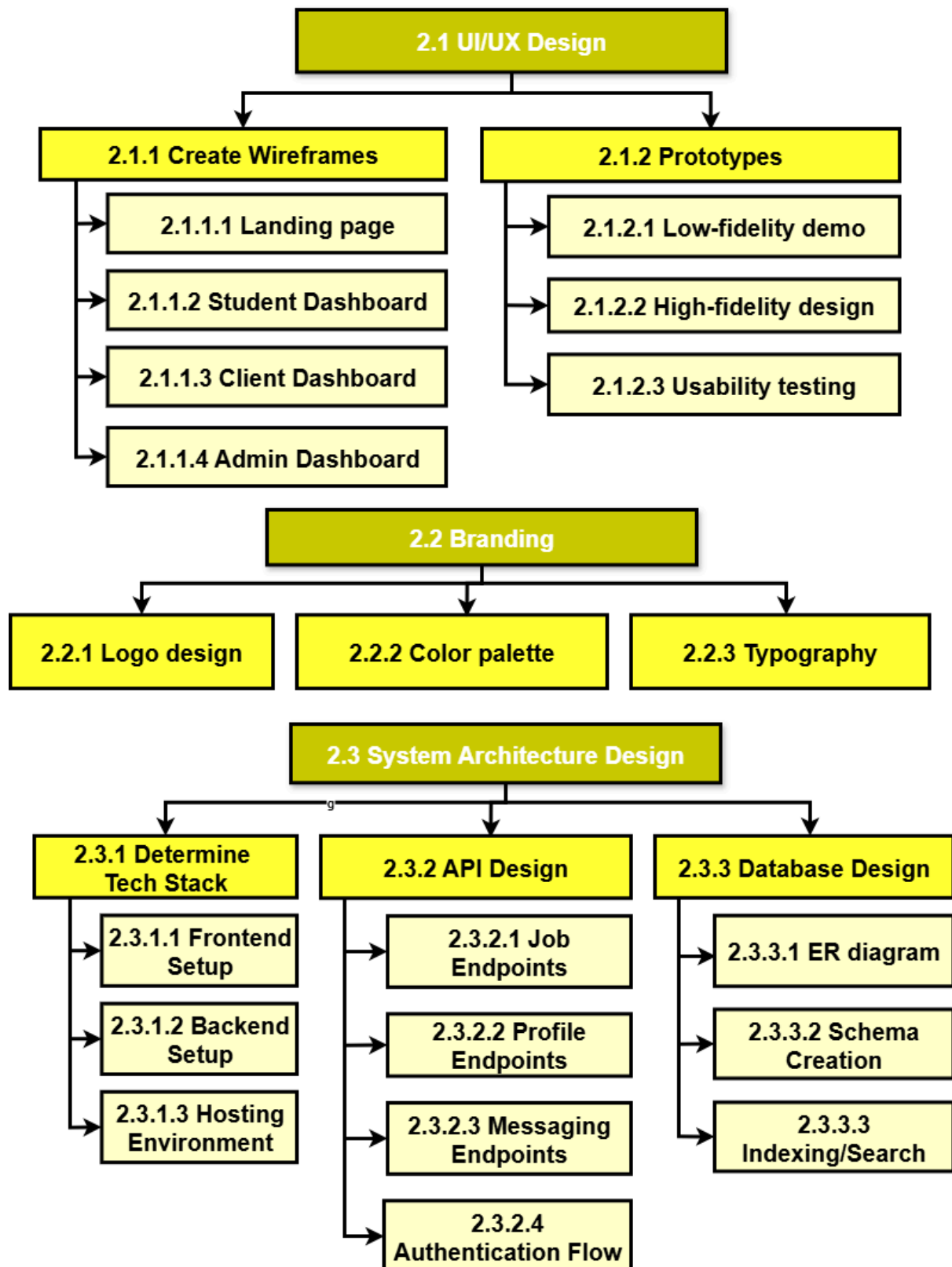
WBS (Layer 0 - Layer 2)

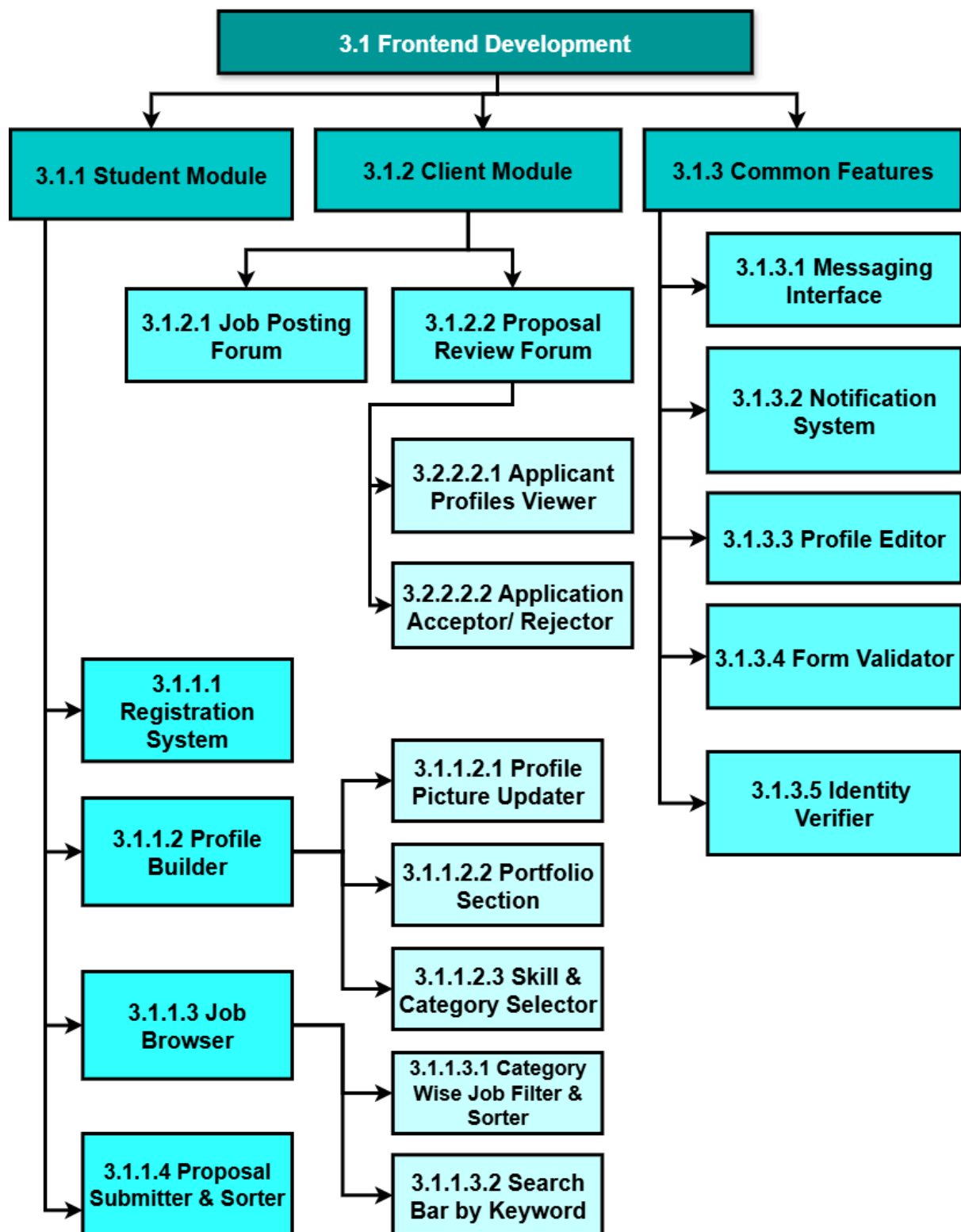


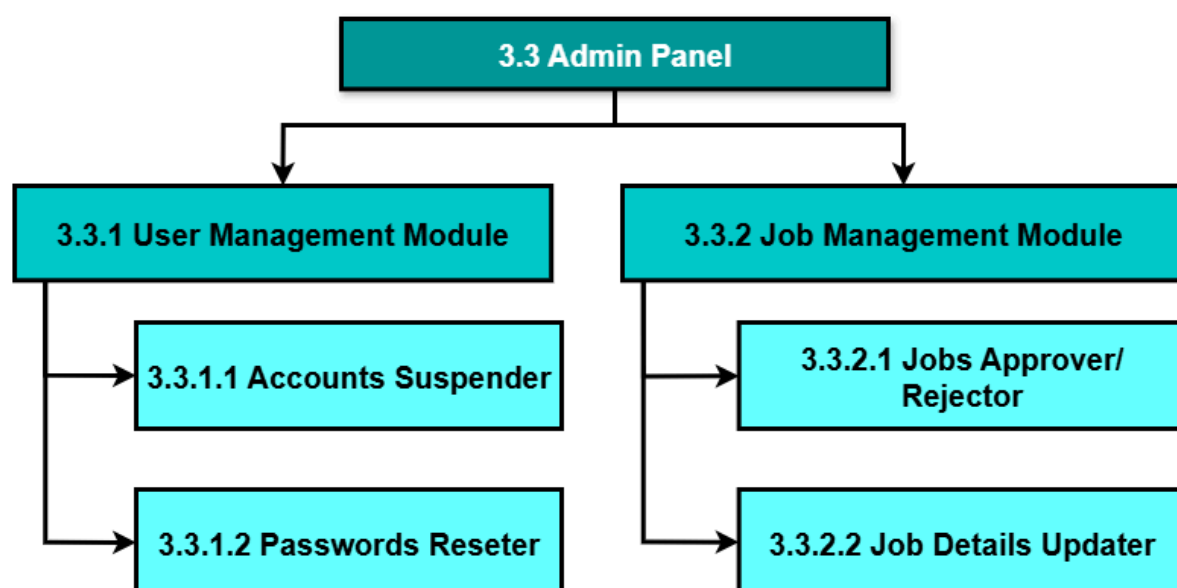
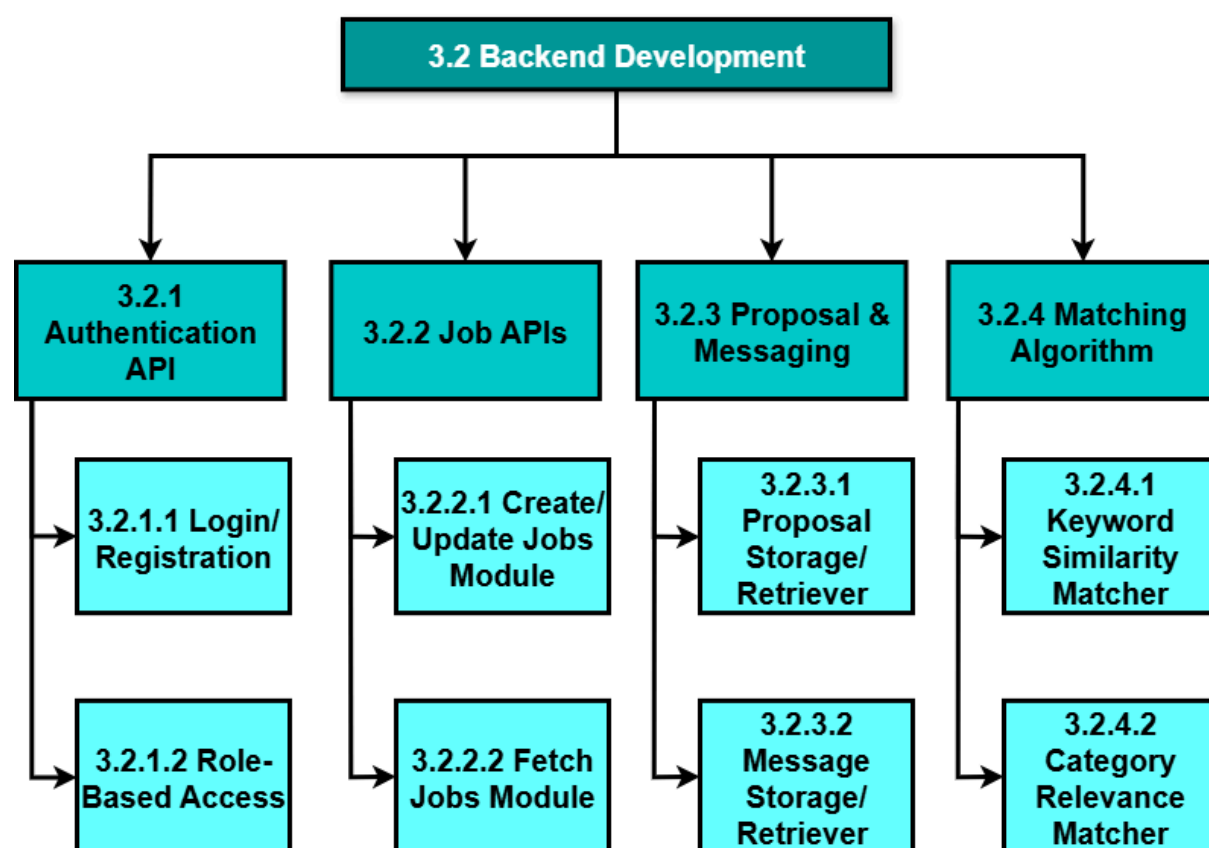
Detailed WBS From Layer 2

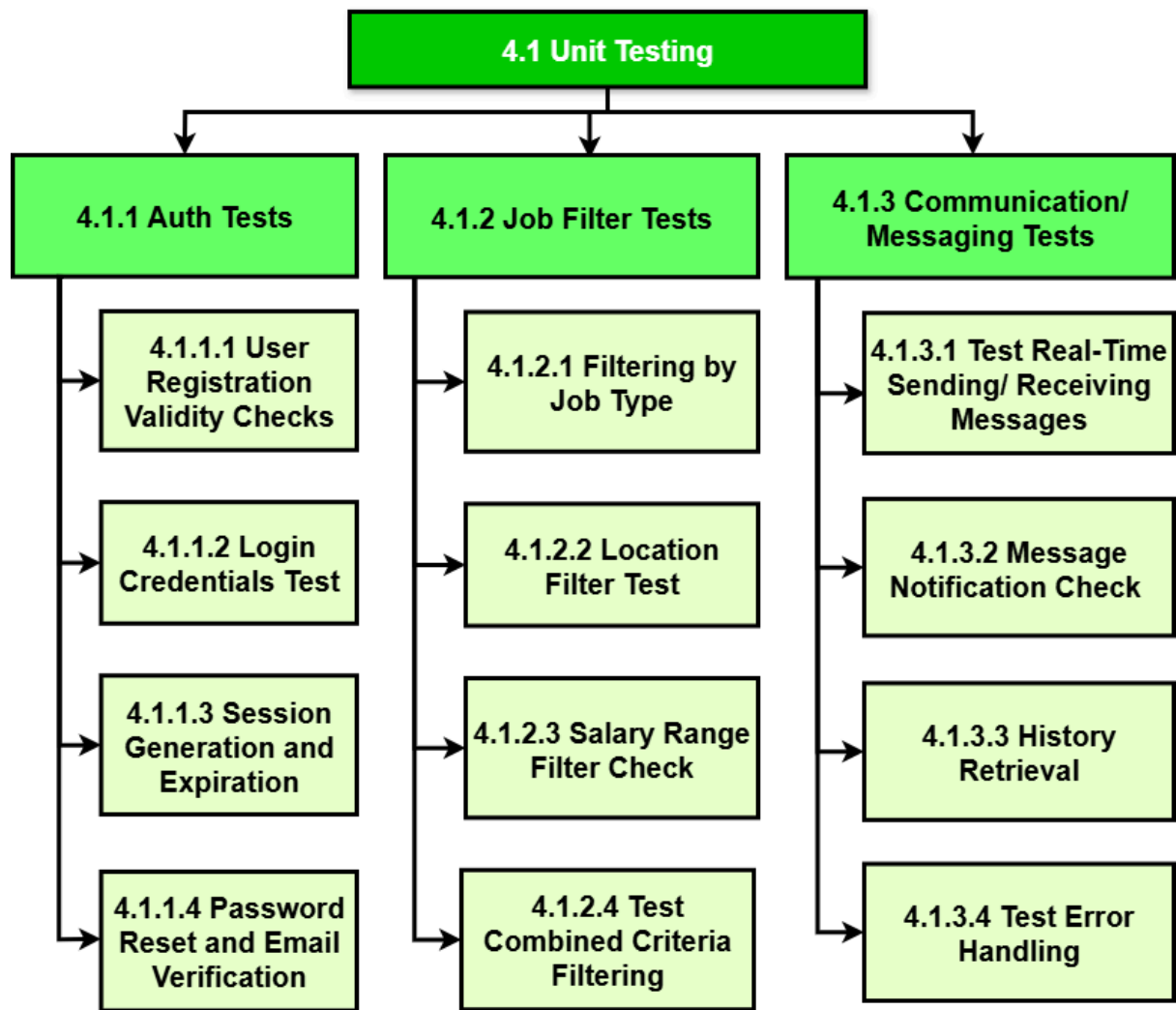


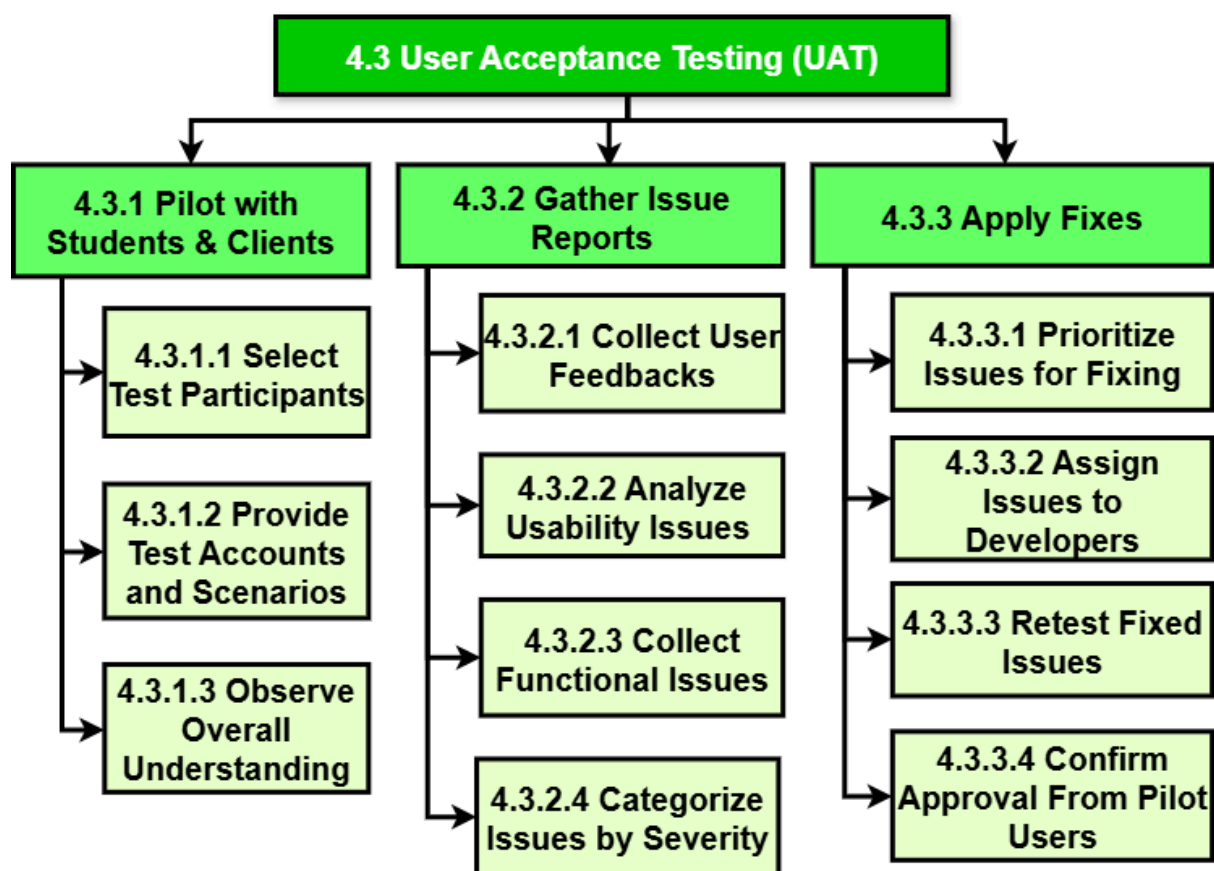
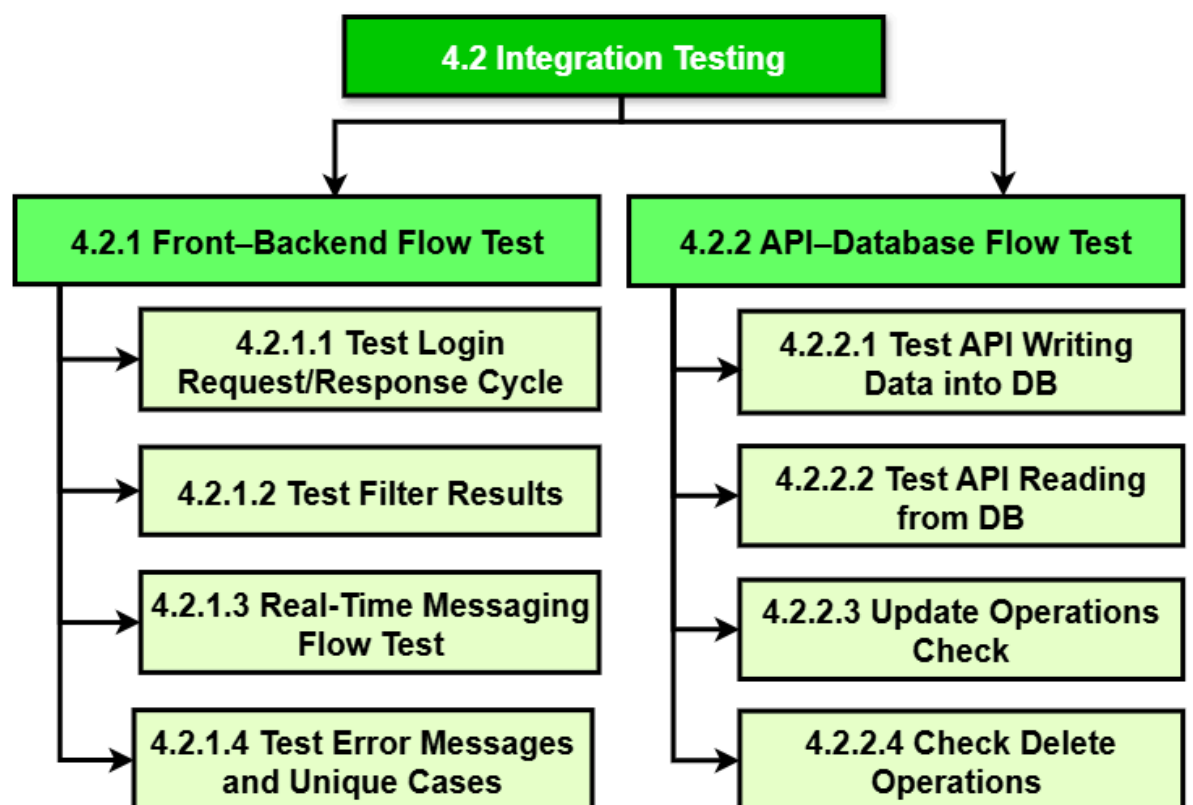


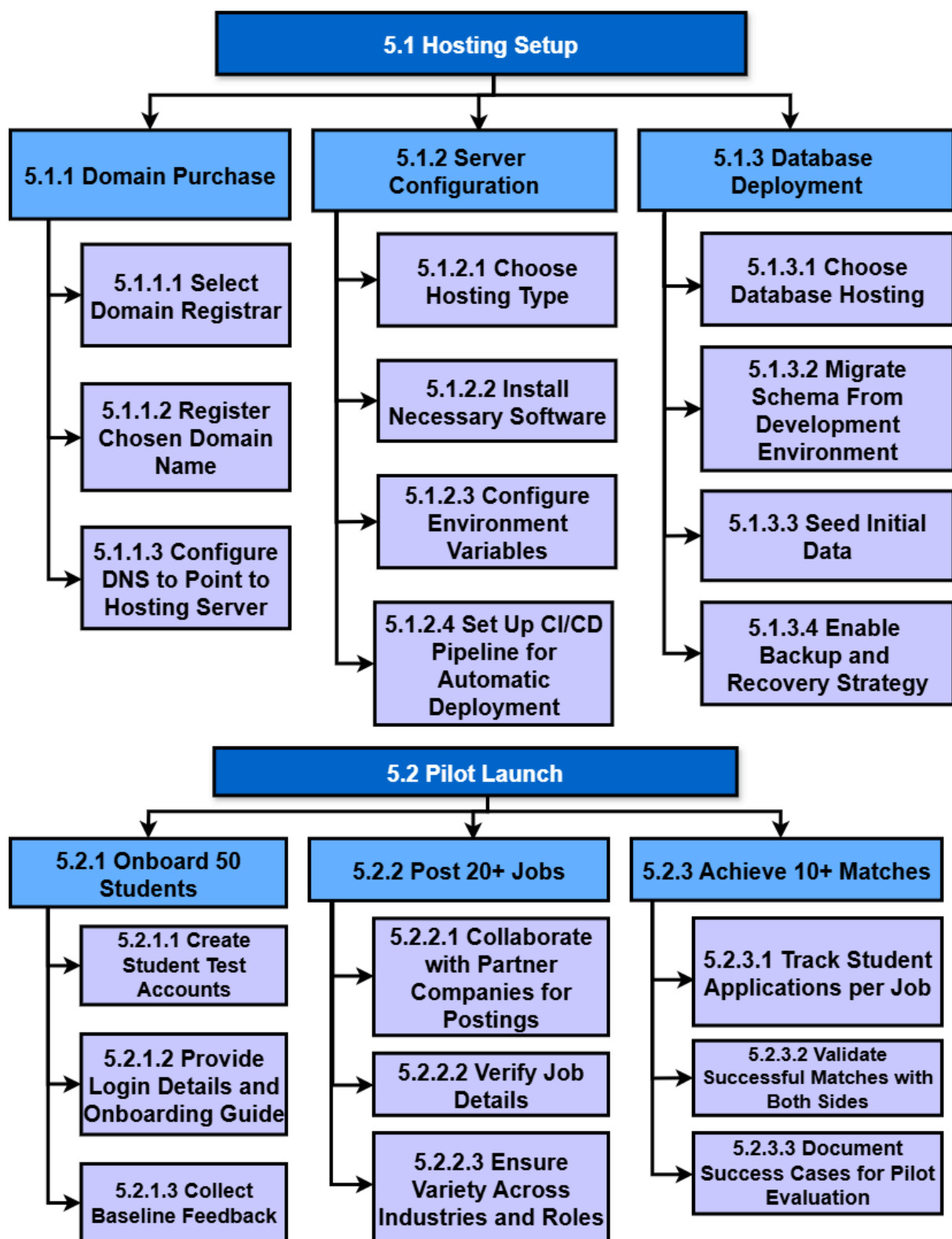


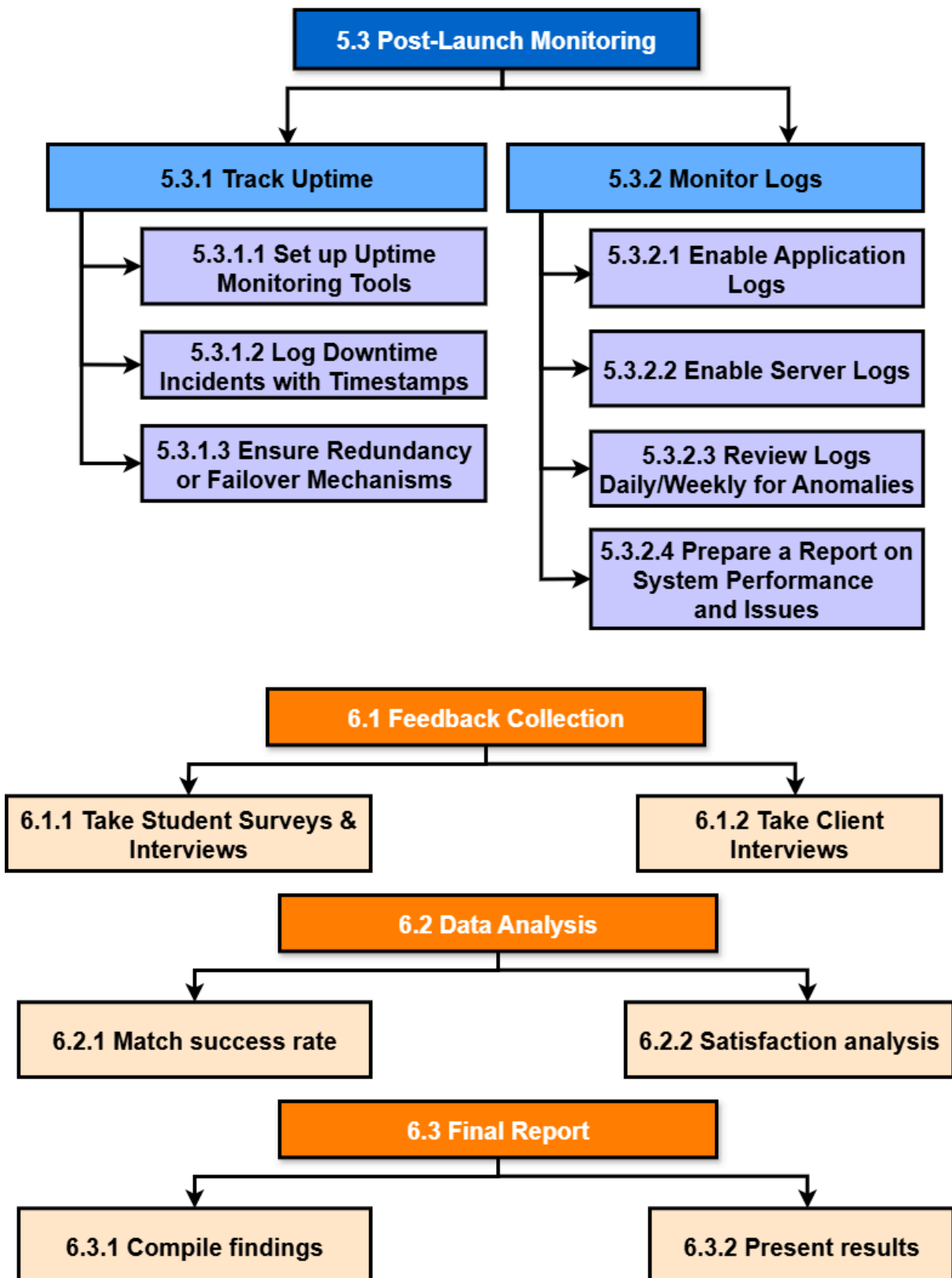


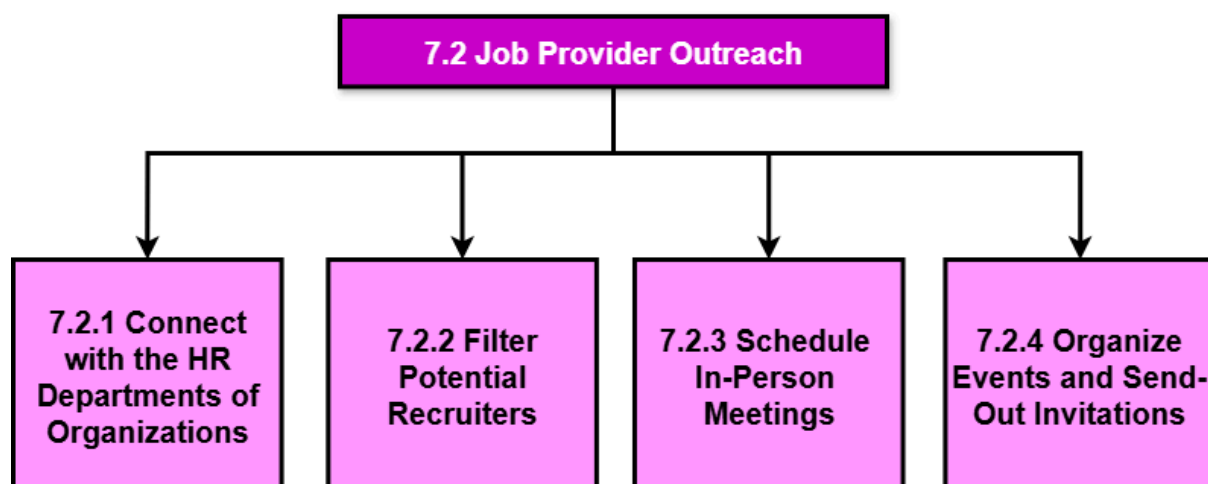
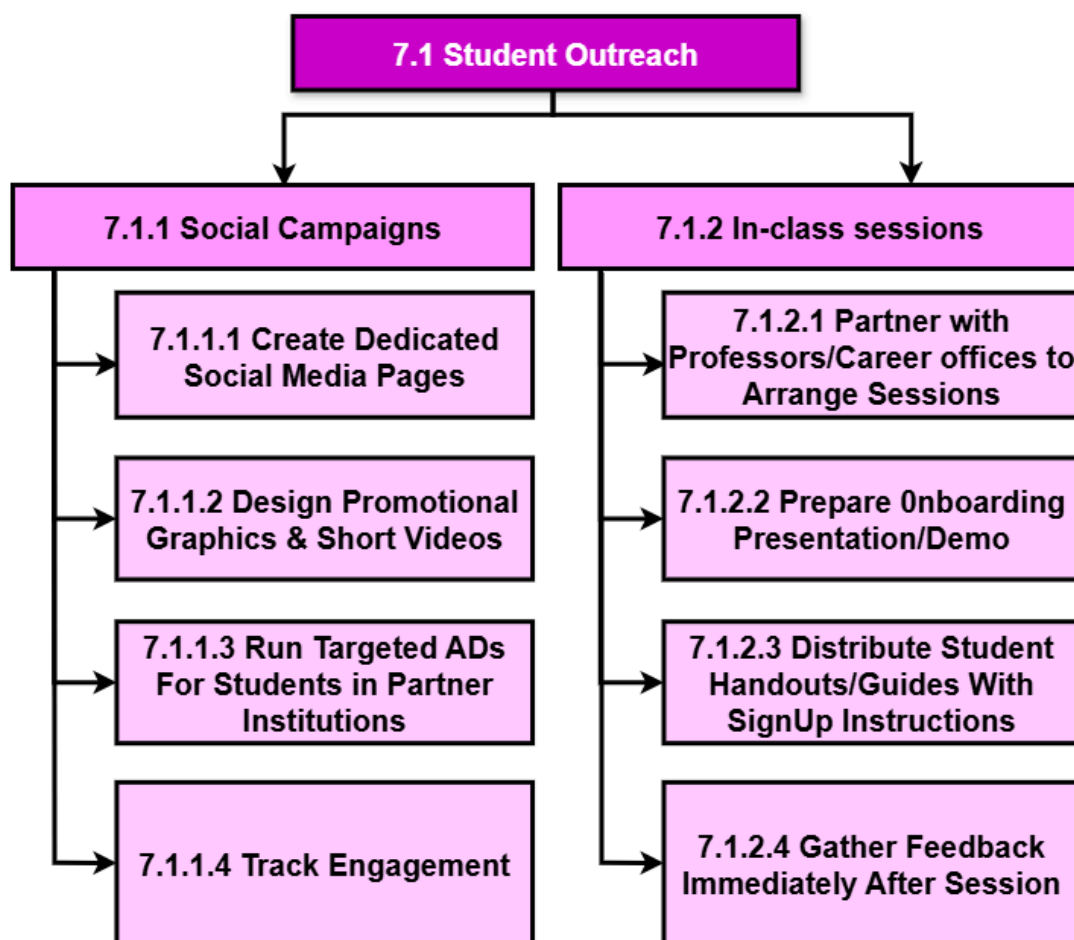


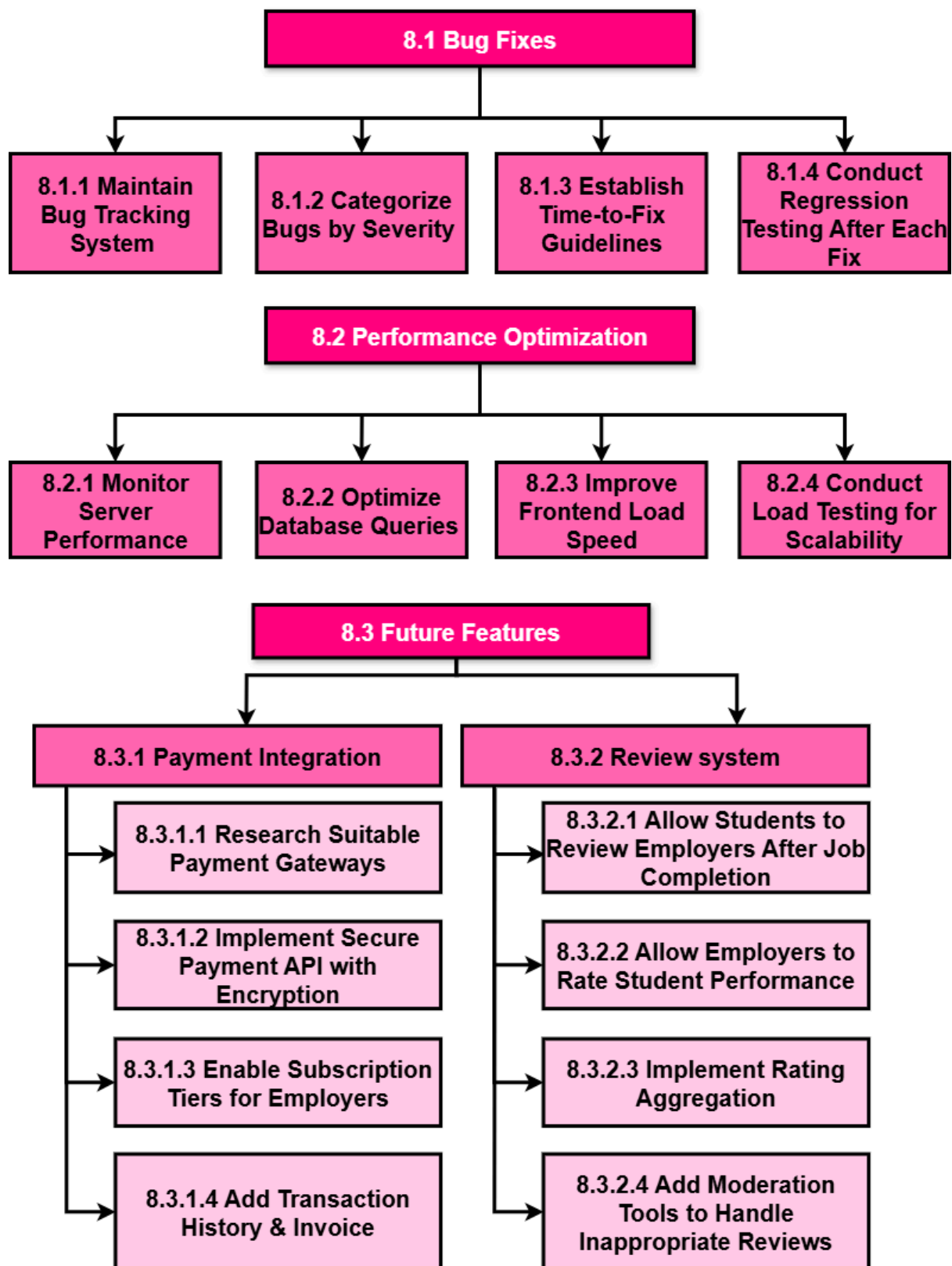










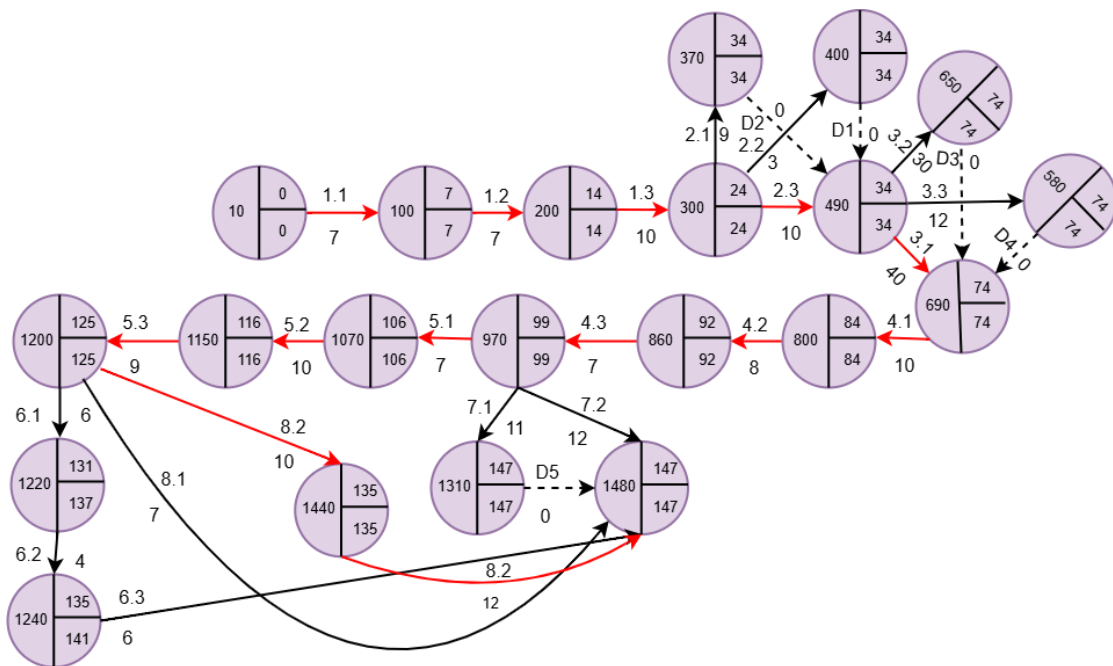


2. Network Diagram

For 0.0 SkillMatch

Node	Predecessor	Duration
1.0	None	24 days
1.1	None	7 days
1.2	1.1	7 days
1.3	1.2	10 days
2.0	1.3	10 days
2.1	1.3	9 days
2.2	1.3	3 days
2.3	1.3	10 days
3.0	2.0	40 days
3.1	2.0	40 days
3.2	2.0	30 days
3.3	2.0	12 days
4.0	3.0	25 days
4.1	3.0	10 days
4.2	4.1	8 days
4.3	4.2	7 days
5.0	4.3	26 days
5.1	4.3	7 days
5.2	5.1	10 days
5.3	5.2	9 days

6.0	5.3	16 days
6.1	5.3	6 days
6.2	6.1	4 days
6.3	6.2	6 days
7.0	4.3	12 days
7.1	4.3	11 days
7.2	4.3	12 days
8.0	5.3	22 days
8.1	5.3	7 days
8.2	5.3	10 days
8.3	8.2	12 days

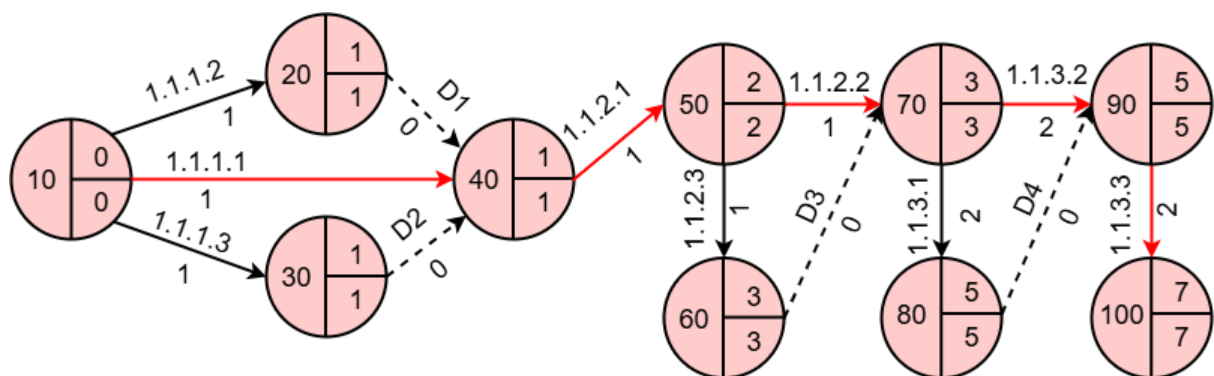


Critical Path: 10 → 100 → 200 → 300 → 400 → 690 → 800 → 860 → 970 → 1070 → 1150 → 1200 → 1440 → 1480

Total Time for whole Project: 7 + 7 + 10 + 10 + 40 + 10 + 8 + 7 + 7 + 10 + 9 + 10 + 12 = 147 days

For 1.1 Project Planning

Node	Predecessor	Duration
1.1.1	None	1 day
└1.1.1.1	None	1 day
└1.1.1.2	None	1 day
└1.1.1.3	None	1 day
1.1.2	1.1.1	2 days
└1.1.2.1	1.1.1	1 day
└1.1.2.2	1.1.2.1	1 day
└1.1.2.3	1.1.2.1	1 day
1.1.3	1.1.2	4 days
└1.1.3.1	1.1.2	2 days
└1.1.3.2	1.1.2	2 days
└1.1.3.3	1.1.3.1,1.1.3.2	2 days

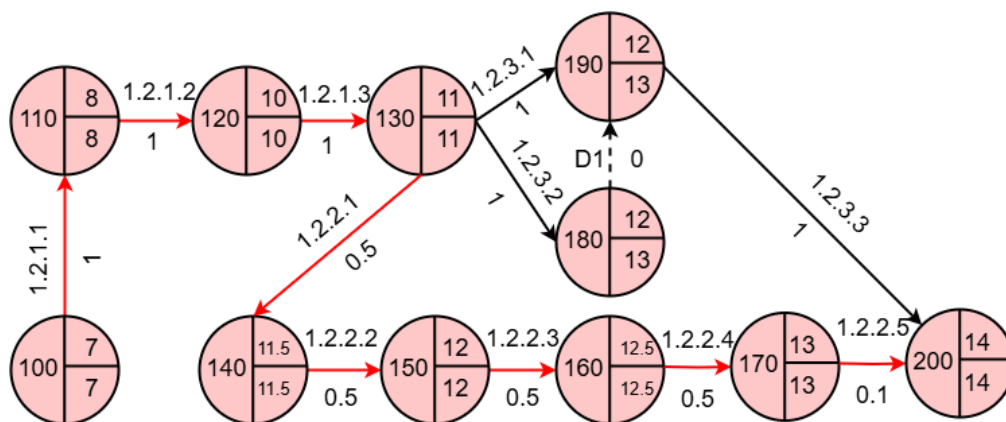


Critical Path: 10 → 40 → 50 → 70 → 90 → 100

Total Time for 1.1: 7 days

For 1.2 Project Initiation

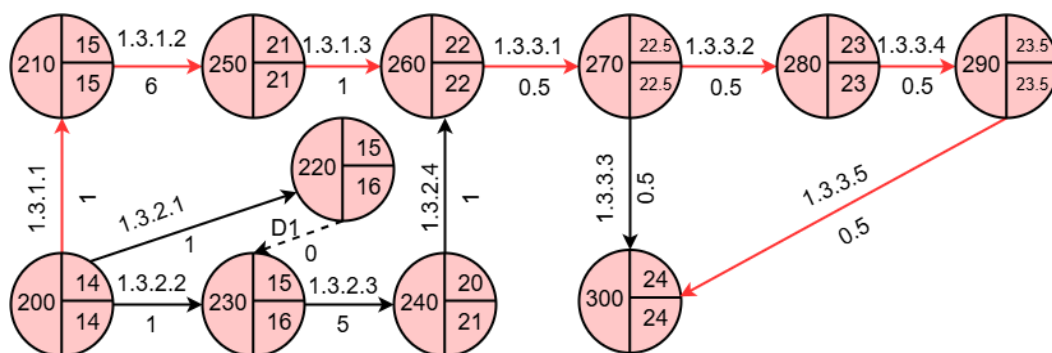
Node	Predecessor	Duration
1.2.1	1.1	4 days
└1.2.1.1	1.1	1 day
└1.2.1.2	1.2.1.1	2 days
└1.2.1.3	1.2.1.2	1 day
1.2.2	1.2.1	3 days
└1.2.2.1	1.2.1	0.5 days
└1.2.2.2	1.2.2.1	0.5 days
└1.2.2.3	1.2.2.2	0.5 days
└1.2.2.4	1.2.2.3	0.5 days
└1.2.2.5	1.2.2.4	1 day
1.2.3	1.2.1	2 days
└1.2.3.1	1.2.1	1 day
└1.2.3.2	1.2.1	1 day
└1.2.3.3	1.2.3.1, 1.2.3.2	1 day



Critical Path: 100 → 110 → 120 → 130 → 140 → 150 → 160 → 170 → 200
Total Time for 1.2: 7 days

For 1.3 Requirement Gathering

Node	Predecessor	Duration
1.3.1	1.2	8 days
└1.3.1.1	1.2	1 day
└1.3.1.2	1.3.1.1	6 days
└1.3.1.3	1.3.1.2	1 day
1.3.2	1.2	8 days
└1.3.2.1	1.2	1 day
└1.3.2.2	1.2	1 day
└1.3.2.3	1.3.2.1, 1.3.2.2	5 days
└1.3.2.4	1.3.2.3	1 day
1.3.3	1.3.1 , 1.3.2	2 days
└1.3.3.1	1.3.1, 1.3.2	0.5 days
└1.3.3.2	1.3.3.1	0.5 days
└1.3.3.3	1.3.3.1	0.5 days
└1.3.3.4	1.3.3.3	0.5 days
└1.3.3.5	1.3.3.4	0.5 days

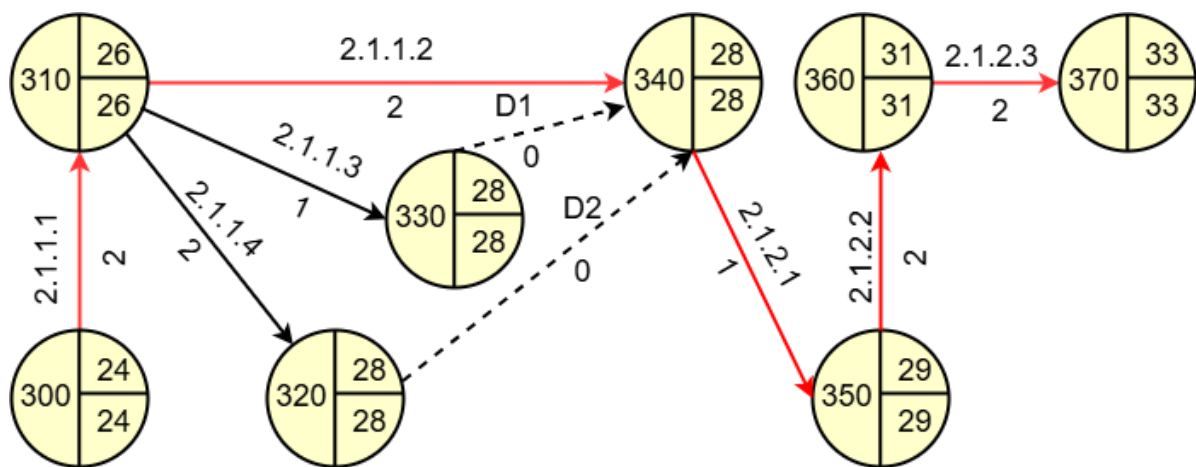


Critical Path: 200 → 210 → 250 → 260 → 270 → 280 → 290 → 300

Total Time for 1.3: 10 days

For 2.1 Ui/UX Design

Node	Predecessor	Duration
2.1.1	1.3	4 days
└2.1.1.1	1.3	2 days
└2.1.1.2	2.1.1.1	2 days
└2.1.1.3	2.1.1.1	1 day
└2.1.1.4	2.1.1.1	2 days
2.1.2	2.1.1	5 days
└2.1.2.1	2.1.1	1 day
└2.1.2.2	2.1.2.1	2 days
└2.1.2.3	2.1.2.2	2 days

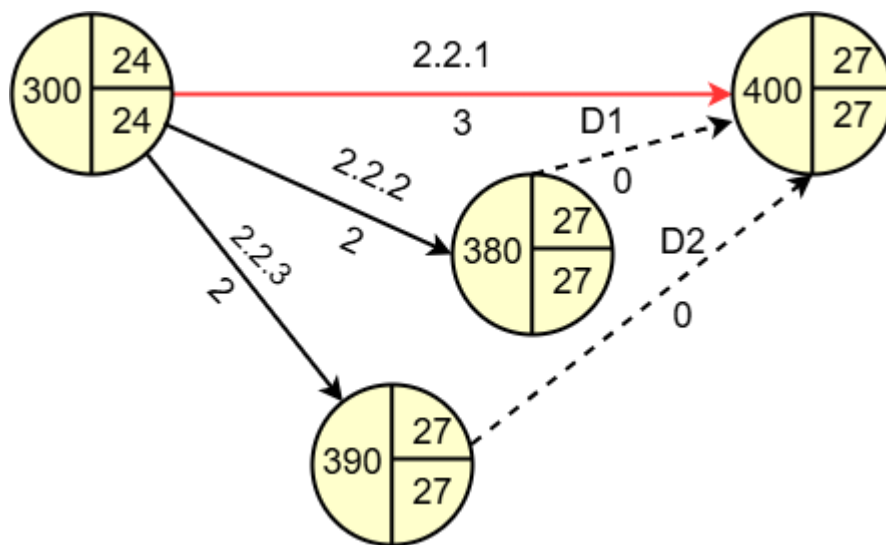


Critical Path: 300 → 310 → 340 → 350 → 360 → 370

Total Time for 2.1: 9 days

For 2.2 Branding

Node	Predecessor	Duration
2.2	1.3	3 days
└2.2.1	1.3	3 days
└2.2.2	1.3	2 days
└2.2.3	1.3	2 days

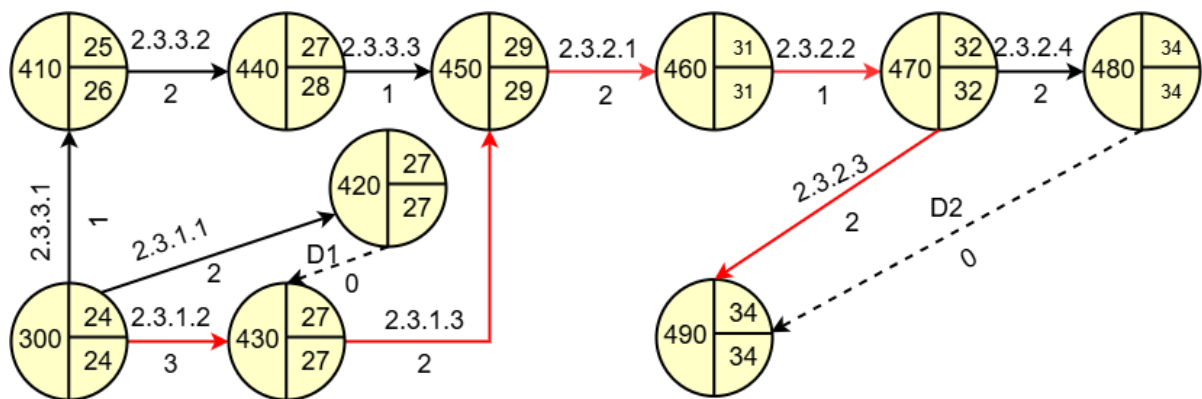


Critical Path: 300 → 400

Total Time for 2.2: 3 days

For 2.3 System Architecture

Node	Predecessor	Duration
2.3.1	1.3	5 days
└2.3.1.1	1.3	2 days
└2.3.1.2	1.3	3 days
└2.3.1.3	2.3.1.2	2 days
2.3.2	2.3.1	5 days
└2.3.2.1	2.3.1	2 day
└2.3.2.2	2.3.2.1	1 day
└2.3.2.3	2.3.2.2	2 days
└2.3.2.4	2.3.2.2	2 day
2.3.3	1.3	4 days
└2.3.3.1	1.3	1 day
└2.3.3.2	2.3.3.1	2 days
└2.3.3.3	2.3.3.2	1 day

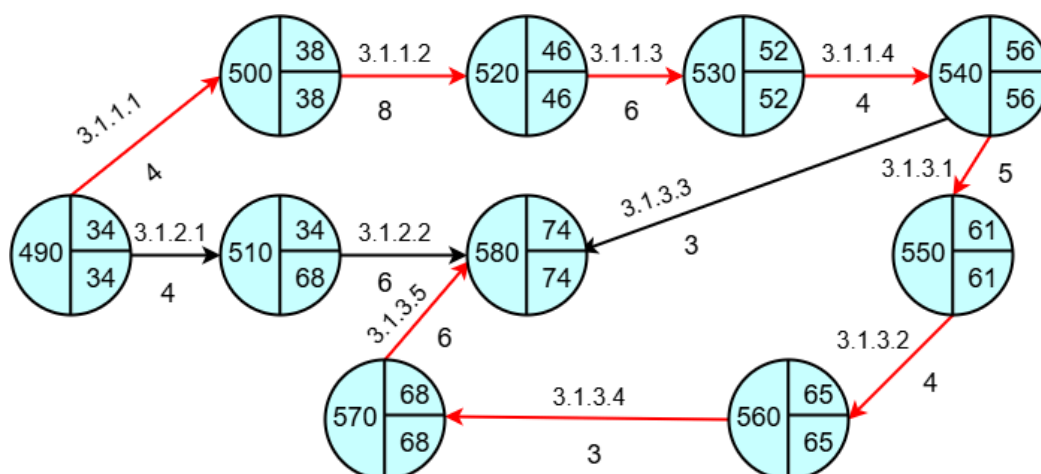


Critical Path: 300 → 430 → 450 → 460 → 470 → 490

Total Time for 2.3: 10 days

For 3.1 Front Development

Node	Predecessor	Duration
3.1.1	2.3	22 days
└3.1.1.1	2.3	4 days
└3.1.1.2	3.1.1.1	8 days
└3.1.1.3	3.1.1.2	6 days
└3.1.1.4	3.1.1.3	4 days
3.1.2	2.3	10 days
└3.1.2.1	2.3	4 days
└3.1.2.2	3.1.2.1	6 days
3.1.3	3.1.1.4	18 days
└3.1.3.1	3.1.1.4	5 days
└3.1.3.2	3.1.3.1	4 days
└3.1.3.3	3.1.1.4	3 days
└3.1.3.4	3.1.3.2	3 days
└3.1.3.5	3.1.3.4	6 days

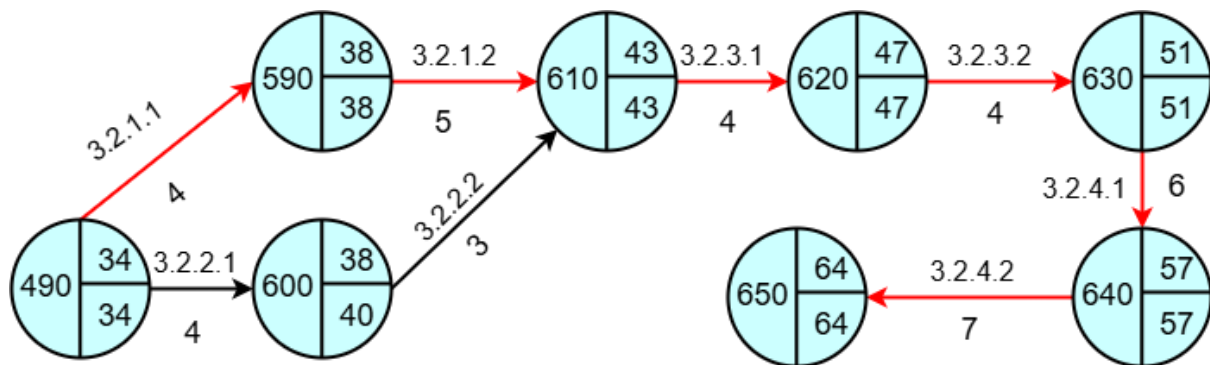


Critical Path: 490 → 500 → 520 → 530 → 540 → 550 → 560 → 570 → 580

Total Time for 3.1: 40 days

For 3.2 Backend Development

Node	Predecessor	Duration
3.2.1	2.3	9 days
└3.2.1.1	2.3	4 days
└3.2.1.2	3.2.1.1	5 days
3.2.2	3.2.1	7 days
└3.2.2.1	3.2.1	4 days
└3.2.2.2	3.2.2	3 days
3.2.3	3.2.1, 3.2.2	8 days
└3.2.3.1	3.2.1, 3.2.2	4 days
└3.2.3.2	3.2.3.1	4 days
3.2.4	3.2.3.2	13 days
└3.2.4.1	3.2.3.2	6 days
└3.2.4.2	3.2.4.1	7 days

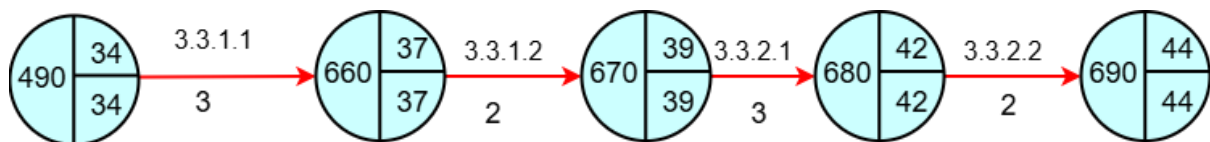


Critical Path: 490 → 500 → 520 → 530 → 540 → 550 → 560 → 570 → 580

Total Time for 3.1: 30 days

For 3.3 Admin Panal

Node	Predecessor	Duration
3.3.1	2.3	5 days
└─3.3.1.1	2.3	3 days
└─3.3.1.2	3.3.1.1	2 days
3.3.2	3.3.1.2	5 days
└─3.3.2.1	3.3.1.2	3 days
└─3.3.2.2	3.3.2.1	2 days



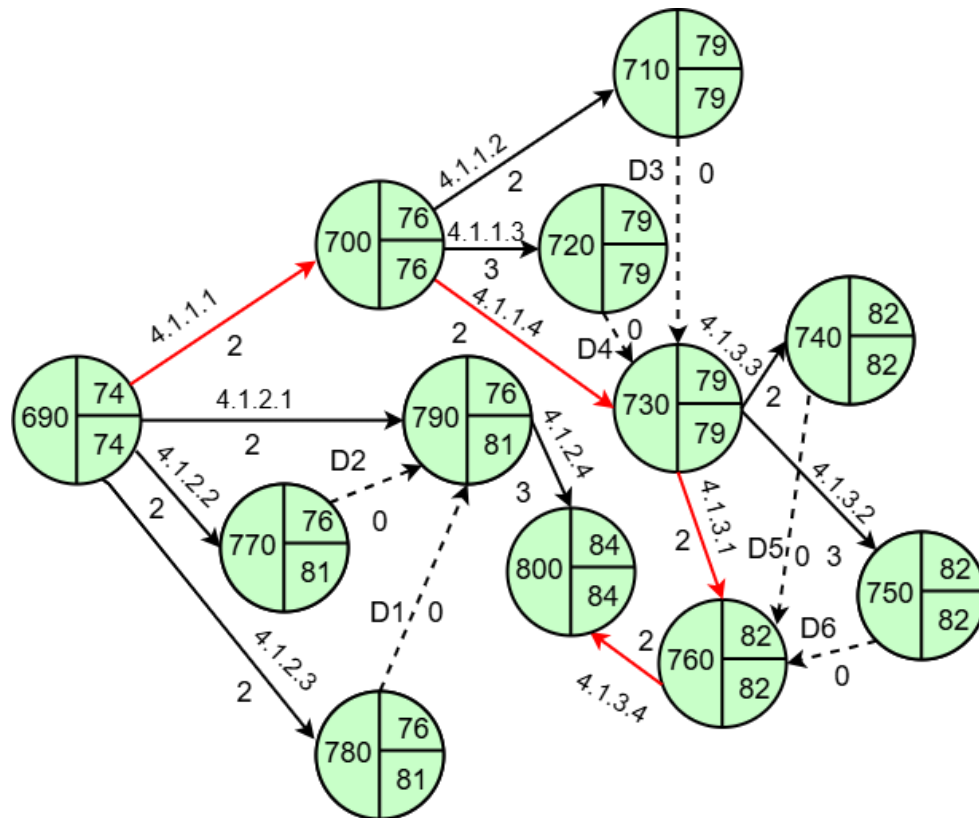
Critical Path: 490 → 500 → 520 → 530 → 540 → 550 → 560 → 570 → 580

Total Time for 3.1: 30 days

For 4.1 unit Testing

Node	Predecessor	Duration
4.1.1	3.0	5 days
└─4.1.1.1	3.0	2 days
└─4.1.1.2	4.1.1.1	2 days
└─4.1.1.3	4.1.1.1	3 days
└─4.1.1.4	4.1.1.1	2 days
4.1.2	3.0	5 days
└─4.1.2.1	3.0	2 days
└─4.1.2.2	3.0	2 days
└─4.1.2.3	3.0	2 days
└─4.1.2.4	4.1.2.1, 4.1.2.2, 4.1.2.3	3 days

4.1.3	4.1.1.4	5 days
└─4.1.3.1	4.1.1.4	2 days
└─4.1.3.2	4.1.3.1	3 days
└─4.1.3.3	4.1.1.4	2 days
└─4.1.3.4	4.1.1.4	2 days

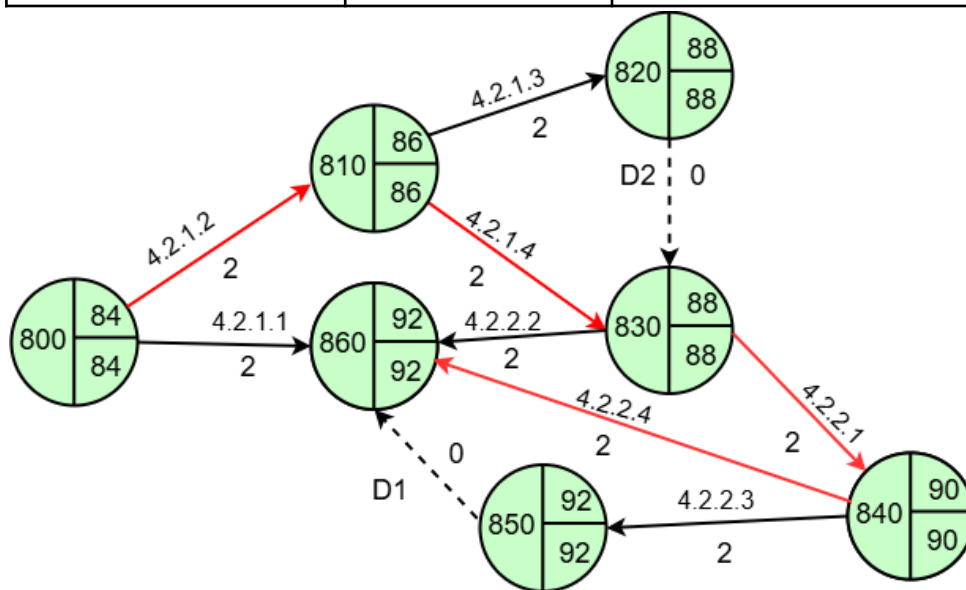


Critical Path: 690 → 700 → 730 → 760 → 800

Total Time for 4.1: 10 days

For 4.2 Integration Testing

Node	Predecessor	Duration
4.2.1	4.1	4 days
├─4.2.1.1	4.1	2 days
├─4.2.1.2	4.1	2 days
├─4.2.1.3	4.2.1.2	2 days
└─4.2.1.4	4.2.1.2	2 days
4.2.2	4.2.1.4, 4.2.1.3	4 days
├─4.2.2.1	4.2.1.4, 4.2.1.3	2 days
├─4.2.2.2	4.2.1.4, 4.2.1.3	2 days
├─4.2.2.3	4.2.2.1	2 days
└─4.2.2.4	4.2.2.1	2 days

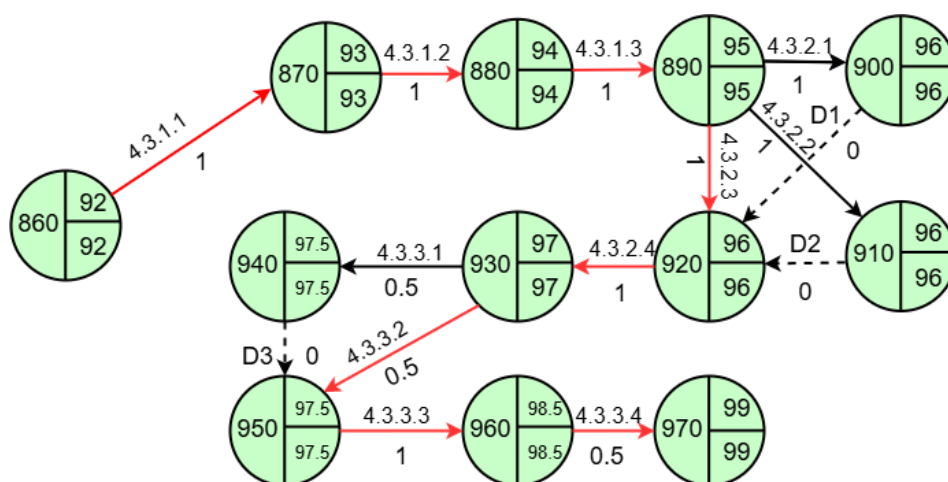


Critical Path: 800 → 810 → 830 → 840 → 860

Total Time for 4.2: 8 days

For 4.3 User Acceptance Testing

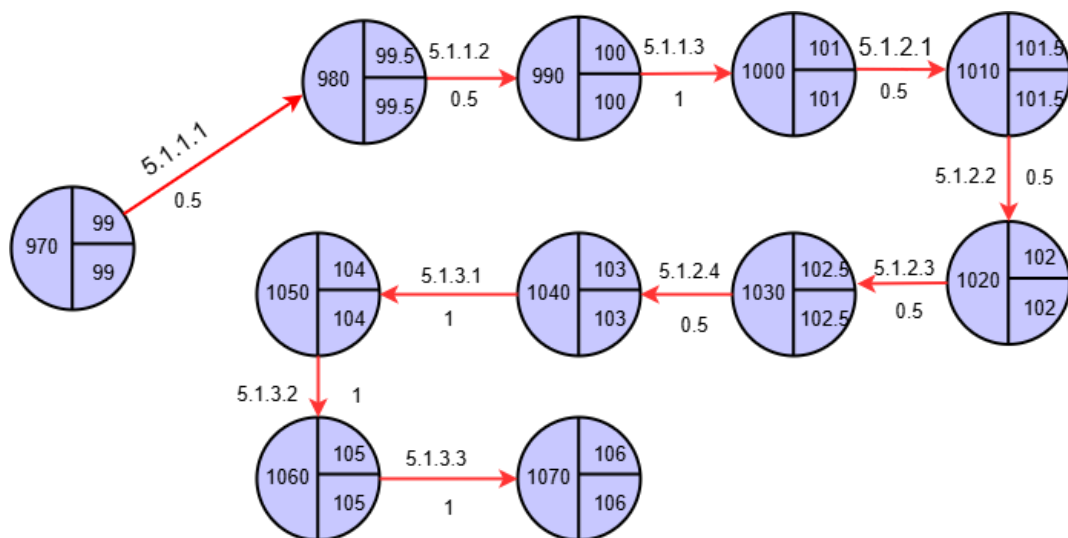
Node	Predecessor	Duration
4.3.1	4.2	3 days
└─4.3.1.1	4.2	1 day
└─4.3.1.2	4.3.1.1	1 day
└─4.3.1.3	4.3.1.2	1 day
4.3.2	4.3.1.3	2 days
└─4.3.2.1	4.3.1.3	1 day
└─4.3.2.2	4.3.1.3	1 day
└─4.3.2.3	4.3.1.3	1 day
└─4.3.2.4	4.3.2.1, 4.3.2.2, 4.3.2.3	1 day
4.3.3	4.3.2.4	2 days
└─4.3.3.1	4.3.2.4	0.5 day
└─4.3.3.2	4.3.2.4	0.5 day
└─4.3.3.3	4.3.3.1, 4.3.3.2	1 day
└─4.3.3.4	4.3.3.3	0.5 day



Critical Path: 860 → 870 → 870 → 890 → 920 → 930 → 950 → 960 → 970
Total Time for 4.3: 7 days

For 5.1 Hosting Setup

Node	Predecessor	Duration
5.1.1	4.3	2 days
└5.1.1.1	4.3	0.5 day
└5.1.1.2	5.1.1.1	0.5 day
└5.1.1.3	5.1.1.2	1 day
5.1.2	5.1.1	2 days
└5.1.2.1	5.1.1	0.5 day
└5.1.2.2	5.1.2.1	0.5 day
└5.1.2.3	5.1.2.2	0.5 day
└5.1.2.4	5.1.2.3	0.5 day
5.1.3	5.1.2	3 days
└5.1.3.1	5.1.2	1 day
└5.1.3.2	5.1.3.1	1 day
└5.1.3.3	5.1.3.2	1 day

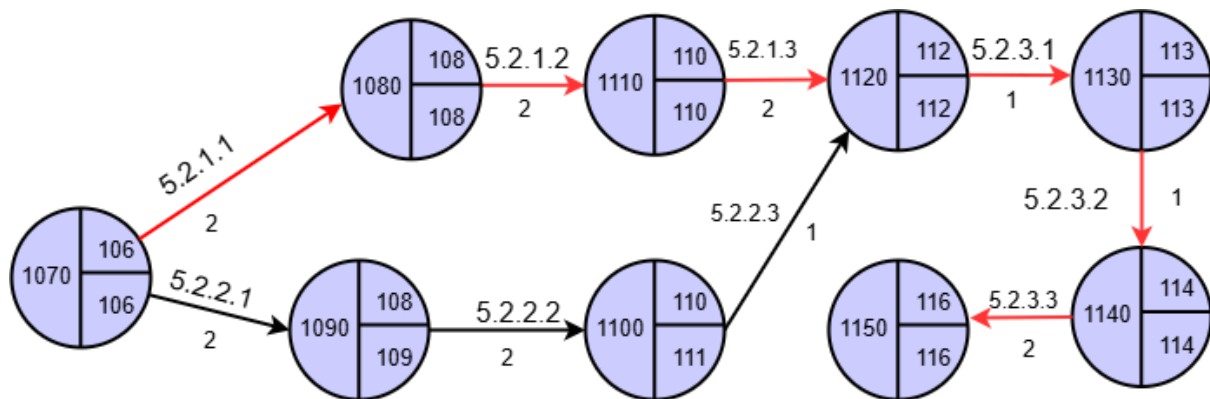


Critical Path: 970 → 980 → 990 → 1000 → 1010 → 1020 → 1030 → 1040 → 1050 → 1060 → 1070

Total Time for 5.1: 7 days

For 5.2 Pilot Launch

Node	Predecessor	Duration
5.2.1	5.1	6 days
└5.2.1.1	5.1	2 days
└5.2.1.2	5.2.1.1	2 days
└5.2.1.3	5.2.1.2	2 days
5.2.2	5.1	5 days
└5.2.2.1	5.1	2 days
└5.2.2.2	5.2.2.1	2 days
└5.2.2.3	5.2.2.2	1 days
5.2.3	5.2.1, 5.2.2	4 days
└5.2.3.1	5.2.1, 5.2.2	1 days
└5.2.3.2	5.2.3.1	1 days
└5.2.3.3	5.2.3.2	2 days

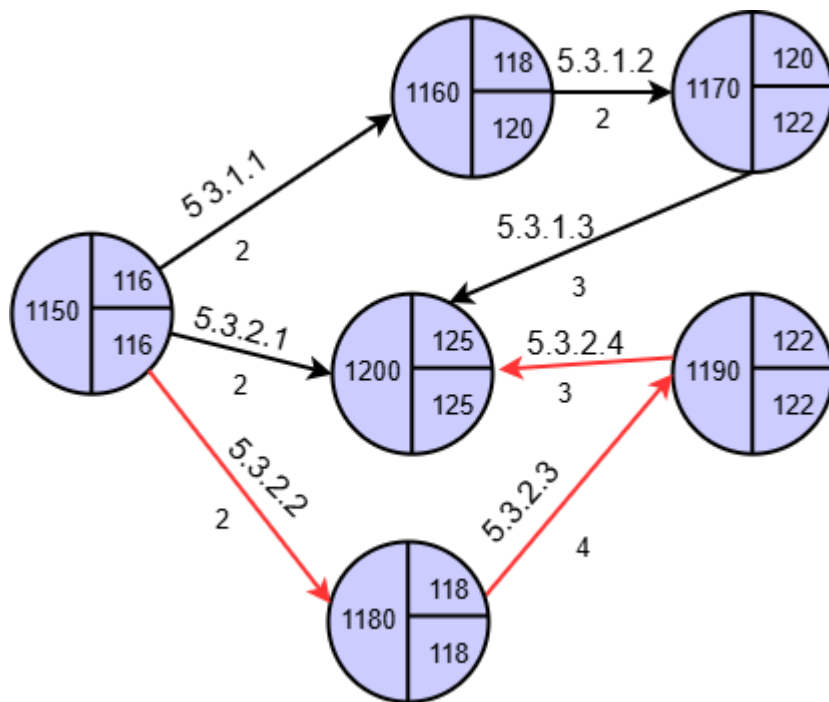


Critical Path: 1070 → 1080 → 1110 → 1120 → 1130 → 1140 → 1150

Total Time for 5.2: 10 days

For 5.3 Post Launch Monitoring

Node	Predecessor	Duration
5.3.1	5.2	5 days
└5.3.1.1	5.2	2 days
└5.3.1.2	5.3.1.1	2 days
└5.3.1.3	5.3.1.2	3 days
5.3.2	5.2	9 days
└5.3.2.1	5.2	2 days
└5.3.2.2	5.2	2 days
└5.3.2.3	5.3.2.2	4 days
└5.3.2.4	5.3.2.3	3 days

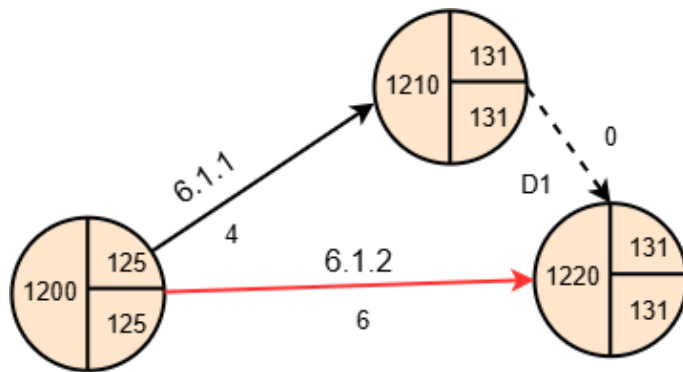


Critical Path: 1150 → 1180 → 1190 → 1200

Total Time for 5.3: 9 days

For 6.1 Feedback Collection

Node	Predecessor	Duration
6.1.1	5.3	4 days
6.1.2	5.3	6 days

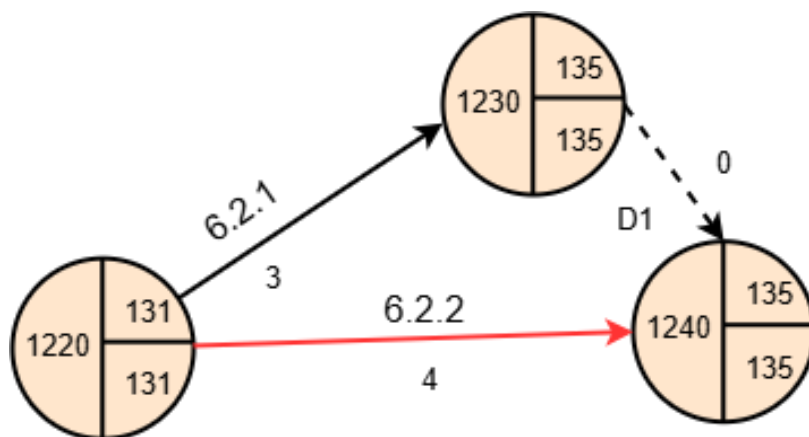


Critical Path: 1200 → 1220

Total Time for 6.1: 6 days

For 6.2 Data Analysis

Node	Predecessor	Duration
6.2.1	6.1	3 days
6.2.2	6.1	4 days

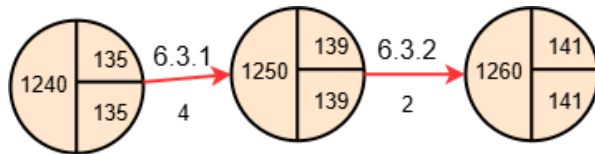


Critical Path: 1220 → 1240

Total Time for 6.2: 4 days

For 6.3 Final Repot

Node	Predecessor	Duration
6.3.1	6.2	4 days
6.3.2	6.3.1	2 days

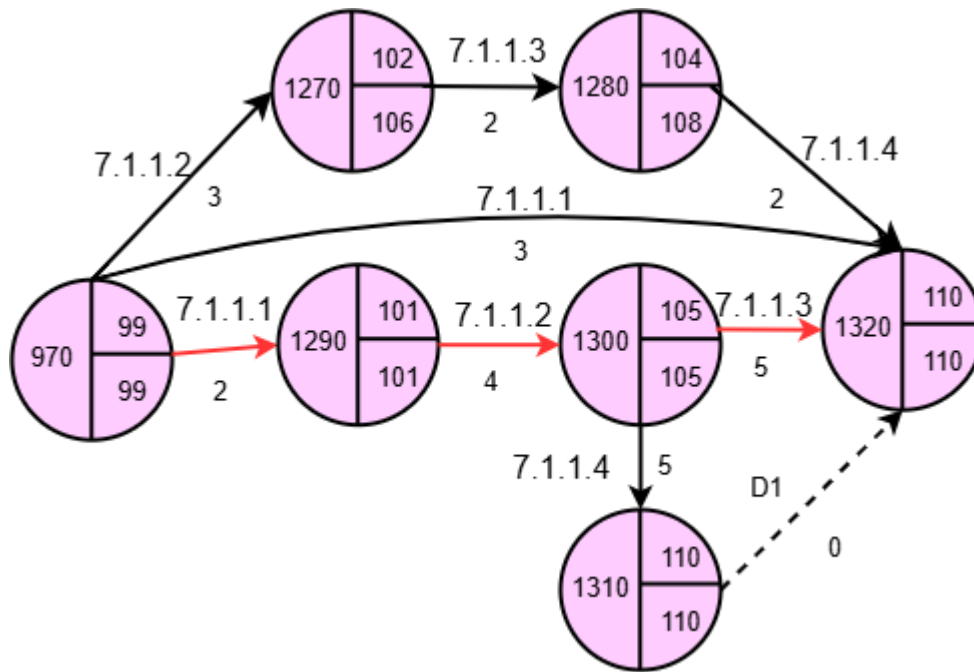


Critical Path: 1240 → 1250 → 1260

Total Time for 6.3: 6 days

For 7.1 Student Outreach

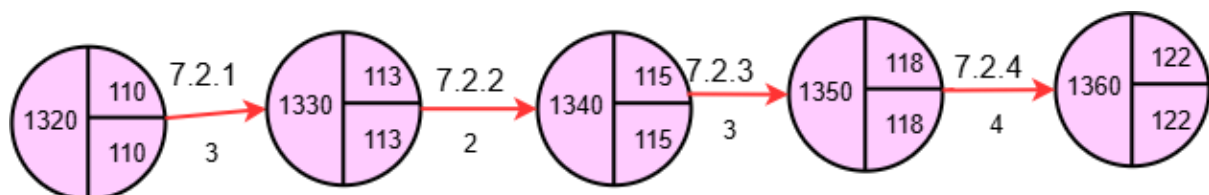
Node	Predecessor	Duration
7.1.1	4.3	11 days
└7.1.1.1	4.3	2 days
└7.1.1.2	7.1.1.1	4 days
└7.1.1.3	7.1.1.2	5 days
└7.1.1.4	7.1.1.2	5 days
7.1.2	4.3	7 days
└7.1.2.1	4.3	3 days
└7.1.2.2	4.3	3 days
└7.1.2.3	7.1.2.2	2 days
└7.1.2.4	7.1.2.3	2 days



Critical Path: 970 → 1290 → 1300 → 1320
Total Time for 7.1: 11 days

For 7.2 Job Provider Outreach

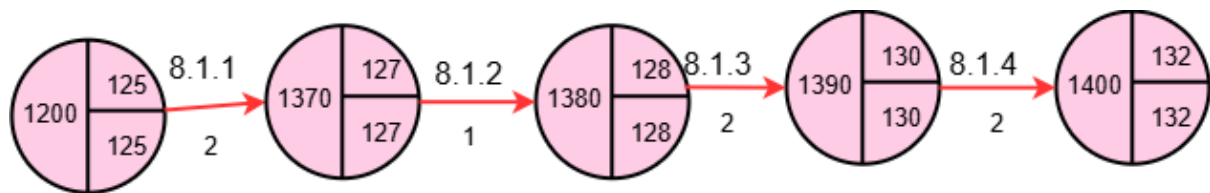
Node	Predecessor	Duration
7.2.1	4.3	3 days
7.2.2	7.2.1	2 days
7.2.3	7.2.2	3 days
7.2.4	7.2.3	4 days



Critical Path: 1320 → 1330 → 1340 → 1350 → 1360
Total Time for 7.2: 12 days

For 8.1 Bug Fixes

Node	Predecessor	Duration
8.1.1	5.0	2 days
8.1.2	8.1.1	1 day
8.1.3	8.1.2	2 days
8.1.4	8.1.3	2 days

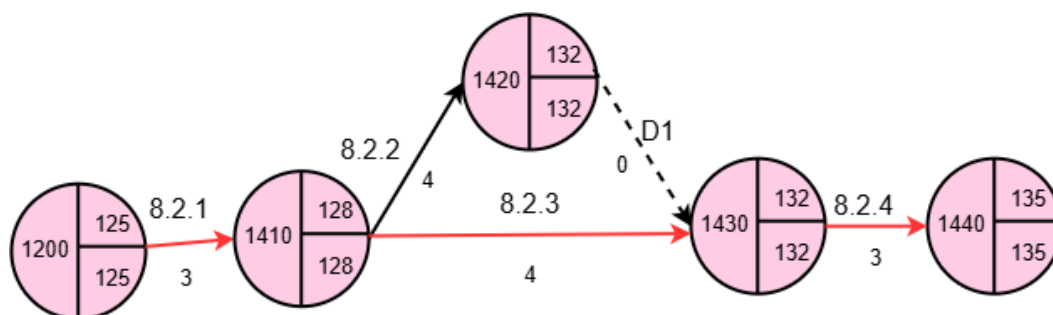


Critical Path: 1200 → 1370 → 1380 → 1390 → 1400

Total Time for 8.1: 7 days

For 8.2 Performance Optimization

Node	Predecessor	Duration
8.2.1	5.0	3 days
8.2.2	8.2.1	4 days
8.2.3	8.2.1	4 days
8.2.4	8.2.2, 8.2.3	3 days

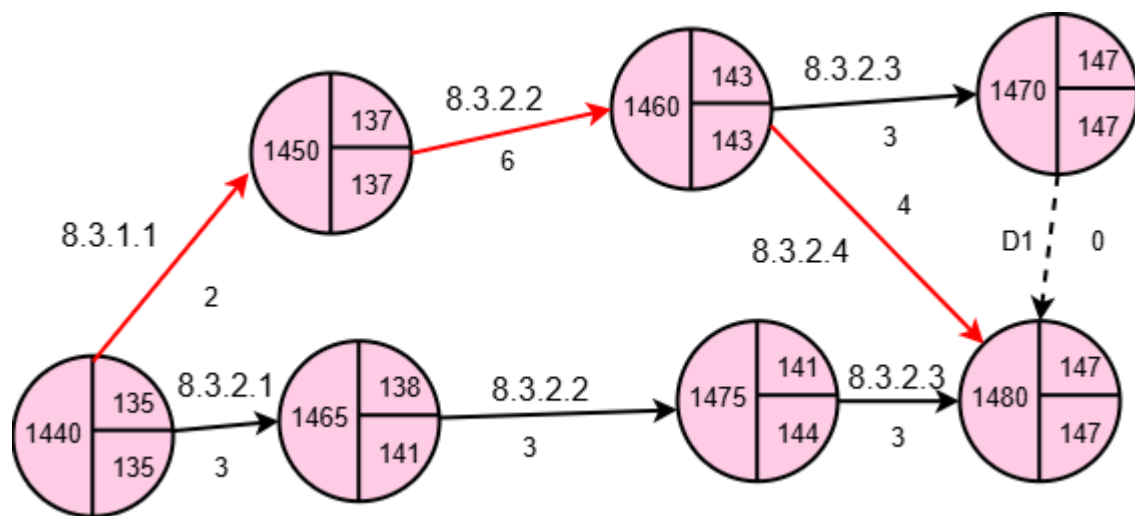


Critical Path: 1200 → 1410 → 1430 → 1440

Total Time for 8.2: 10 days

For 8.3 Future Features

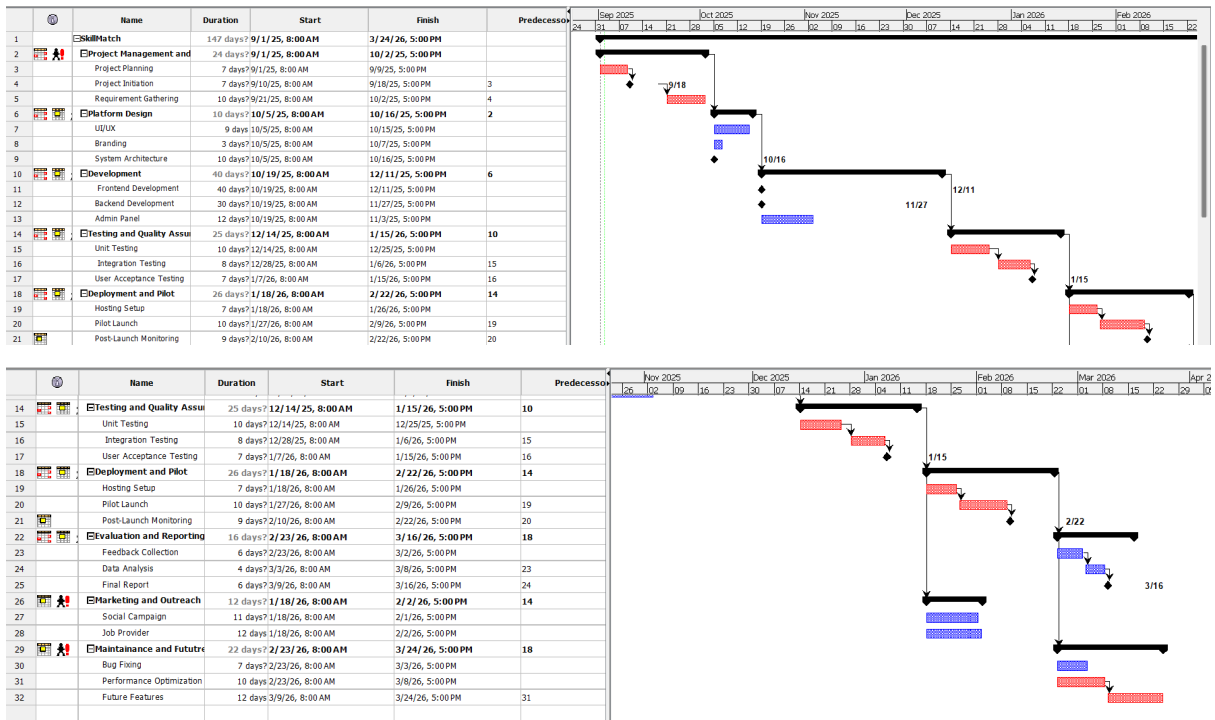
Node	Predecessor	Duration
8.3.1	8.2.1	12 days
8.3.1.1	8.2.1	2 days
8.3.1.2	8.3.1.1	6 days
8.3.1.3	8.3.1.2	4 days
8.3.1.4	8.3.1.2	3 days
8.3.2	8.2.1	9 days
8.3.2.1	8.2.1	3 days
8.3.2.2	8.3.2.1	3 days
8.3.2.3	8.3.2.2	3 days



Critical Path: 1440 → 1450 → 1460 → 1470

Total Time for 8.3: 12 days

3. Gantt Chart



5. Costing

1. Rationale Behind the Software Costing Estimation Model

1.1 Rationale for Choosing Dynamic (COCOMO Model)

When estimating software costs, two main approaches are available: **Static models** and **Dynamic models**.

i. **Static models** (single-variable or multi-variable) provide fixed estimates. Single-variable models rely on only one input factor such as Lines of Code (LOC), while multi-variable models include additional factors like complexity, team experience, and environment. However, static models do not adapt to changes in project scope, team capability, or technology. They tend to oversimplify complex projects and often underestimate effort.

ii. **Dynamic models** (like COCOMO) treat all variables as interdependent. They account for project attributes such as size, complexity, tools, and team experience. This makes them more accurate for projects where requirements evolve and where teams have mixed levels of expertise.

Since *SkillMatch* is not a trivial class project but a structured initiative with planning, design, development, testing, deployment, and maintenance, the **dynamic COCOMO model** is the most suitable. It captures variability in staff skills, technology use, and project challenges—factors that a static model would miss.

1.2 Rationale for Semi-Detached from Dynamic COCOMO Model

The COCOMO model divides projects into three categories:

- i. **Organic**: Small, simple projects, experienced teams, stable requirements.
- ii. **Semi-Detached**: Medium-sized projects, teams with mixed experience, moderate complexity, some unfamiliar aspects.
- iii. **Embedded**: Large, highly complex, hardware-coupled, or regulation-bound systems.

The *SkillMatch* project clearly fits **Semi-Detached** because:

- **Project Size**: Medium (~5 months, ~160+ days), covering design, development, testing, and deployment.
- **Team Composition**: A mix of relatively inexperienced student developers supported by experienced instructors/supervisors.

- **Complexity:** Standard modules (CRUD features, dashboards, messaging) combined with more challenging parts (matching algorithm, deployment pipeline).
- **Uncertainty Level:** Moderate — some modules are straightforward, while others involve higher risk (scaling, user acceptance testing, integrations).
- **Technology Stack:** Uses common web technologies (React, Firebase/PHP, MySQL), which are not trivial but also not subject to heavy regulation or hardware constraints.

For these reasons, the **Semi-Detached mode of the Dynamic COCOMO model** is the best choice for estimating effort, time, and cost for the *SkillMatch* platform.

2. Software Costing

Software costing involves estimating and determining the expenses associated with developing, deploying, and maintaining software throughout its lifecycle. Properly understanding and managing costs is crucial for budgeting, resource allocation, and project planning.

The detailed costing for *SkillMatch* is broken down into **10 key components**:

1. Development Costs
2. Infrastructure Costs
3. Testing Costs
4. Deployment Costs
5. Maintenance Costs
6. Project Management Costs
7. Licenses & Subscriptions
8. Documentation Costs
9. Overhead Costs
10. Contingency & Risk

(Detailed breakdown follows...)

COCOMO (Semi-Detached) Calculation

From the Gantt chart, the project schedule length is:

Development Time (T_{DEV}) = 147 days

Converting to months:

$$\begin{aligned} T_{DEV} &= 147 \div 30 \\ &\approx \mathbf{4.90 \text{ months}} \end{aligned}$$

1 . Effort (Person-Months)

For the semi-detached mode, development time is related to effort as:

$$\begin{aligned} T_{DEV} &= 2.5 \times (Effort)^{0.35} \\ \Rightarrow \mathbf{Effort} &= (T_{DEV} \div 2.5)^{(1 \div 0.35)} \\ &= (4.90 \div 2.5)^{(2.857)} \\ &\approx \mathbf{(1.96)^{(2.857)}} \\ &\approx \mathbf{6.84 \text{ Person-Months (PM)}} \end{aligned}$$

2. Average Manning (Staffing)

$$\begin{aligned} \mathbf{Average Manning} &= Effort \div T_{DEV} \\ &= 6.84 \div 4.90 \\ &\approx \mathbf{1.40 \text{ Persons}} \end{aligned}$$

This implies on average 1–2 **FTE (Full-Time Equivalent)** persons are required throughout the 147-day schedule.

3. Project Size (KLOC)

The semi-detached COCOMO effort relation is:

$$Effort = 3.0 \times (KLOC)^{1.12}$$

$$\Rightarrow KLOC = (Effort \div 3.0)^{(1 \div 1.12)}$$

$$=(6.84 \div 3.0)^{(0.893)}$$

$$\approx (2.28)^{(0.893)}$$

$$\approx 2.09 \text{ KLOC}$$

$$\approx 2,090 \text{ LOC}$$

4. Productivity

$$Productivity = LOC \div Effort$$

$$Productivity = 2,090 \div 6.84$$

$$\approx 305 \text{ LOC/PM}$$

Final COCOMO Results (Semi-Detached)

- **Effort:** $\approx 6.84 \text{ PM}$
- **Average Manning:** $\approx 1.40 \text{ Persons}$
- **Project Size:** $\approx 2.09 \text{ KLOC}$ ($\approx 2,090 \text{ LOC}$)
- **Productivity:** $\approx 305 \text{ LOC/PM}$

Reconciling with Actual Role Plan

Although the COCOMO model estimates an average staffing of **~1.4 FTE (Full-Time Equivalent)**, the project plan identifies **13 named roles** (Project Manager, System Analyst, UX Designer, 2×Frontend Developers, 2×Backend Developers, QA Engineers, Deployment Engineer, Marketing, and Documentation).

- **Core contributors (full-time in most weeks):** ~1 Frontend + 1 Backend Developer (≈ 1.0 – 1.5 FTE, Full-Time Equivalent).
- **Specialist contributors (short, phase-specific):**
 - Project Manager — continuous but light (10–20%), heavier at milestones.
 - System Analyst — mainly in early planning (Weeks 1–3).
 - UX Designer — design phase (Weeks 1–4), occasional tweaks later.
 - Frontend/Backend — during peak build (Nov–Dec).
 - QA Engineers — during late build & testing (Dec–Jan).
 - Deployment Engineer — during release (Jan–Feb).
 - Marketing/Outreach — during launch preparation (Jan–Feb).
 - Documentation Writer — during the reporting & user guide phase.

Interpretation:

- The **COCOMO estimate (6.84 PM, ~1.4 FTE, Full-Time Equivalent)** represents the *average sustained workload*.
- In reality, all 13 roles are engaged, but mostly part-time and in their respective phases.
- Specialized staff are brought in only when needed, while core developers provide the continuous workload.

Important Note for Costing

Even though COCOMO's mathematical estimate averages out to ~1.4 FTE (Full-Time Equivalent), the **salary-based costing** must still reflect all roles being allocated for the project period (full monthly salaries). In real budgeting, organizations account for the cost of assigning each specialist role, even if their workload is not evenly distributed across the project timeline.

Considerations:

- Exchange rate: **1 USD = 120 BDT**
- Work hours from Gantt Chart

1) Development Costs

i) Labor Costs

Role	Hourly Rate (USD)	Total (USD)	Total (BDT)
UX Designer	1.89	151.20	18,144
Frontend Dev 1	2.37	1,572.48	188,698
Frontend Dev 2	2.37	873.60	104,832
Frontend Dev 3	2.37	758.40	91,008
Backend Dev 1	3.31	1,906.56	228,787
Backend Dev 2	3.31	1,059.20	127,104
System Analyst	4.73	5,552.48	666,298
Marketing 1	2.84	272.64	32,717
Marketing 2	2.84	272.64	32,717
Subtotal Dev Labor	—	10,166.8	1,490,305

ii) Tool & Technology Costs

Item	Cost (USD)	Cost (BDT)
IDEs/GitHub/design plugins (misc.)	167	20,000

Total Development Costs= 10,333.8 USD / 1,510,305 BDT

2) Infrastructure Costs

i) Hardware Costs

Sub-item	Total (USD)	Total (BDT)
Hardware (PCs/laptops for 13 staff)	5,417	650,000

ii) Software Licenses

Item	Cost (USD)	Cost (BDT)
Domain + VPS hosting (pilot)	83	10,000

Total Infrastructure Costs= 660,000 BDT (5,500 USD)

3) Testing Costs

i) Testing Tools

Item	Cost\ (USD)	Cost (BDT)
Selenium/Postman/Jest (addons, misc.)	250	30,000
Bug tracking (Jira/Trello pilot)	167	20,000

ii) QA Team Costs

Role	Hourly Rate (USD)	Total (USD)	Total (BDT)
Tester 1	2.60	977.60	117,312
Tester 2	2.60	520.00	62,400
QA Subtotal	—	1,497.6	179,712

Total Testing Costs (tools only here) =1,914.6 USD / 229,712 BDT

4) Deployment Costs

i) Deployment Tools

Item	Cost (USD)	Cost (BDT)
Docker/CI/CD setup (configs, runners)	83	10,000

ii) User Training

Item	USD	Cost (BDT)
Onboarding materials & sessions	167	20,000

iii)

Role	Hourly Rate (USD)	Total (USD)	Total (BDT)
Devop	3.79	788.32	94,598

Total Deployment Costs (tools + training) = 1,038.3 USD / 124,598 BDT

5) Maintenance Costs

i) Support & Maintenance

ii) Enhancements

→ Maintenance Costs = 417 USD

=50,000 BDT

6) Project Management Costs

i) Project Manager Costs

Role	Hourly Rate (USD)	Total (USD)	Total (BDT)
Project Manager	5.68	7,942	953,040

ii) Contingency Planning

The **budget buffer** is recorded in **Section 10** to avoid double-counting. The PM's time for planning/monitoring is already included above.

Total Project Management Costs =7,942 USD/ 953,040 BDT

7) License & Subscription Costs (Third-Party)

Item	Cost (USD)	Cost (BDT)
Firebase/Cloud APIs/Storage (pilot)	417	50,000

8) Documentation Costs

Item	USD	Cost (BDT)
Formatting/printing manuals & guides	83.33	10,000

9) Overhead Costs

Facility Costs + General Administrative Costs

Item	Cost (BDT)	USD
Internet, electricity, workspace	100,000	833
Stationery & office support	50,000	417
Overhead total	150,000	1,250

10) Contingency & Risk Management

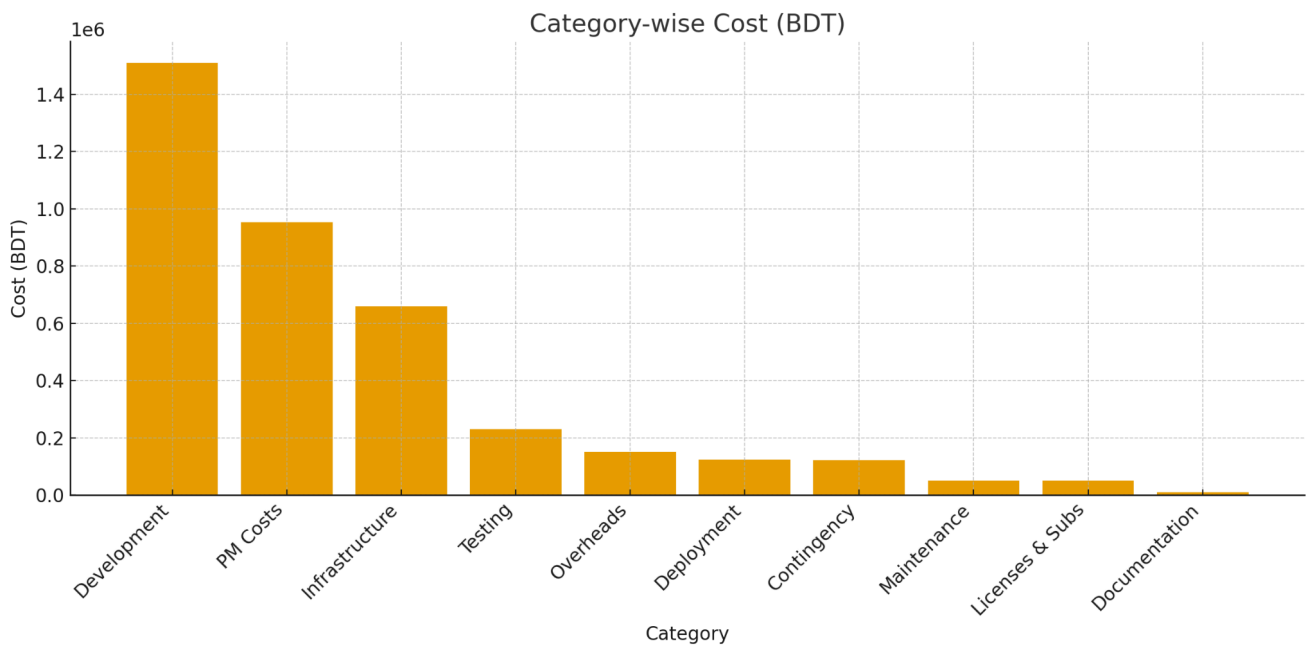
- Defined as **5% of labor** (your salary schedule total).
- Labor total = **10,166.8 + 1,497.6 + 788.3 + 7,942 = 20,394.7 USD**
- **Buffer (5%) = 1,020 USD ($\approx$ 122,400 BDT)**

Basis	Rate	USD	Cost (BDT)
Labor (3,400,000 BDT)	5%	1,020	122,400 BDT

Grand Total Estimated Software Cost

Category	Sub-Item	Sub-Sub Item	Cost (USD)	Cost (BDT)
1. Development	i) Labor Costs	All dev/design/analyst/marketing roles	10,166.8	1,490,305
	ii) Tools & Tech	IDEs, GitHub, plugins, design software	167	20,000
	Total Development	—	10,333.8	1,510,305
2. Infrastructure	i) Hardware	PCs & laptops for 13 staff	5,417	650,000
	ii) Hosting	Domain + VPS	83	10,000
	Total Infra	—	5,500	660,000
3. Testing	i) QA Labor	2 testers (actual hours)	1,497.6	179,712
	ii) Tools	Selenium/Postman/Jest	250	30,000
	iii) Bug Tracking	Jira/Trello	167	20,000
	Total Testing	—	1,914.6	229,712
4. Deployment	i) Labor	Deployment Engineer (actual hours)	788.32	94,598
	ii) Tools	Docker/CI/CD configs	83	10,000
	iii) Training	User onboarding materials	167	20,000
	Total Deployment	—	1,038.3	124,598
5. Maintenance	i) Support & Enhance.		417	50000
6. PM Costs	i) Project Manager	Actual hours from Gantt	7,942	953,040

7. Licenses & Subs	Third-Party Services	Firebase, APIs, storage	417	50,000
8. Documentation	Production	Formatting, printing manuals	83	10,000
9. Overheads	Facility + Admin	Internet, utilities, office support	1,250	150,000
10. Contingency	Risk Buffer (5%)	5% of total labor ($20,394.7 \times 5\%$)	1,020	122,400
GRAND TOTAL	—	—	29,916.7	3,860,055



Conclusion

SkillMatch bridges the gap between students seeking real-world experience and startups needing affordable, skilled talent. By connecting students with meaningful micro-projects, the platform empowers learners, supports businesses, and demonstrates the value of structured project management in delivering an efficient and impactful solution.